From	Our Reference	Tel	Stuttgart-Weilimdorf
SG/PJ-BRM	Rico Schwabe	+49(711) 4009-8854	14.11.2018

Software description for Boost Recuperation Machine

BRM



© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

Content

1 Introduction....3
1.1 Disclaimer....3
1.2 Global Information....4
1.3 Delivered Files....5
2 Changes to previous SW....6
2.1 Functional Updates & Bug-Fixes....6
2.2 Communication-Plug....6
2.3 Flashing....6
3 BRM Hardware-Integration....7
3.1 System-Design....7
3.2 Communication-Interface....8
4 Communication....9
4.1 CAN-Matrix....9
4.2 Transmitted CAN signals by BRM....13
4.3 Received CAN Signals by BRM....16
5 BRM Initialization....20
6 Operation Modes....22
6.1 State Machine....22
6.2 Torque Mode....23
6.3 Speed mode....25
6.4 Voltage mode....28
6.5 Start mode....30
6.6 BRM Concept for Derating, Limitation & Prediction....32
6.7 Prediction signals & EEM calculations....37
7 Diagnostics & Error Memory....42
8 SW Update (Flash)....46
9 Known Bugs....48

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

1 Introduction

1.1 Disclaimer

Document Version: 6.0

This SW description is designed for the BRM SW 320_320_D_01_01.

The described SW supports the following BRM samples:

C3

C4

The present document describes the communication interface as well as the software functionalities underlying the BRM. The aim of this description is to ensure an undisturbed workflow, from the first initiation, to the implementation in customer-specific control strategies, up to a self-contained error analysis.

While working with the BRM, all state of the art safety precautions and advices in the Technical Customer Documentation (TCD or TKU) have to be complied!

The customer is responsible of taking all necessary safety measures on higher level, e.g. test bench or vehicle.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

1.2 Global Information

The BRM (Boost-Recuperation-Machine) consists of two physical components:

- an improved claw pole machine
- and an inverter.

Both components are permanently interconnected (see Image <BRM>).



<BRM>

A control unit is part of the BRM inverter and contains the software to operate the BRM and the communication interface.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

The BRM needs to receive control signals via CAN to be able to operate properly. Therefore the BRM needs to be part of a network. This network contains other control units, for example:

- PT (Powertrain)
- EEM (Electrical Energy Management)
- and the BRM

1.3 Delivered Files

The following content is delivered together with this SW:

	Name	Type
CANoe Configuration	CANoe_OEM	File folder
Vector vflashpack	320320D0101.vflashpack	Vector vFlash Pack&Go
Hex-File	320320D0101_can.hex	HEX File
System Template	320320D0101_OEM.arxml	ARXML File
CAN Database	320320D0101_OEM.dbc	Vector DBC-File
Diagnostic Description	320320D0101_OEM_uds.pdx	PDX File
SW Description	BRM_SW-interface-description_320320D0101_v6.0.pdf	Adobe Acrobat Document

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

2 Changes to previous SW

2.1 Functional Updates & Bug-Fixes

There are several improvements in comparison to the previous BRM SW (320_320_C_07_01):

- BRM initialization at speed
- Runtime improvement

2.2 Communication-Plug

It is strongly recommended to compare the information in the TCD regarding the communication-plug. Especially if the present samples have customer specific modifications the pin assignment can vary.

2.3 Flashing

It is not permitted to flash C- or D-sample SW (i.e. this SW) to a B-sample HW.
It is not permitted to flash B-sample SW (i.e. 320_320_B_04_01) to a C-sample HW.
In case the BRM C-sample was re-flashed with a B-sample SW it will be damaged!
For detailed information see chapter <SW Update (Flash)>.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

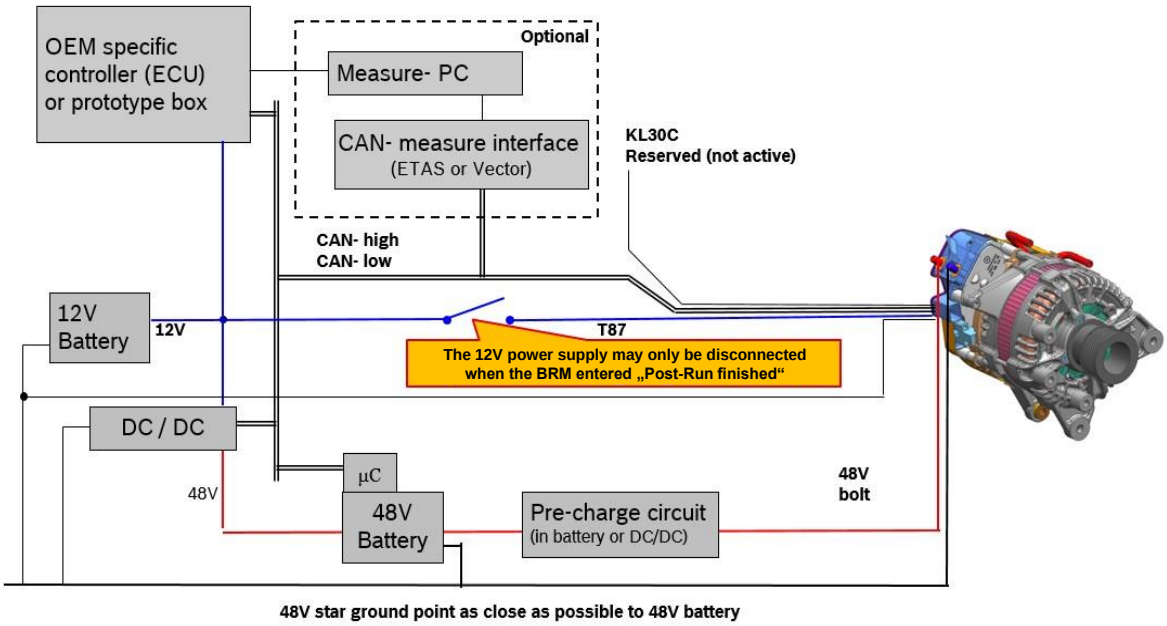
3 BRM Hardware-Integration

3.1 System-Design

For a fault-free and smooth operation, the BRM must be integrated into a component assembly. An example of such an integrated system with its required connections is shown in Image <BRM Integration Example>. The system can be divided into the two electric subsystems 12 V and 48 V.

The electric 12 V system supplies as usual the manufacturer specific components (combustion engine control unit – ECU and several vehicle components) and the BRM control unit (12 V has to be connected via Terminal 87).

The BRM itself (B+, B-), a battery based on 48 V as well as further power loads are connected to the 48 V electric system. A DC/DC converter interconnects between both electric subsystems and enables the electric power and energy transfer beyond the system boundary 12 V / 48 V.



<BRM Integration Example>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

3.2 Communication-Interface

In the C-sample the communicator plug and its pin assignment depend on the customer specific solution. Therefore please refer to the Technical Customer Documentation (TCD/TKU).

For information purpose only, the most typical pin assignments are shown in the images below. In case the communication plug is oriented towards the belt pulley, the pin assignment is shown in the image <Pin assignment - communication plug to belt pulley>. In the other case the pin assignment is shown in the image <Pin assignment - communication plug away from belt pulley>.

To properly operate the BRM, all the listed pins have to be connected according to the illustrations. The communication with the BRM takes place via CAN-bus using an OEM specific controller or a rapid-prototyping system.

Note: As an example the communication plug to belt pulley is shown in image <BRM> in chapter <Introduction / Global Information>.



Pin	Connection	Description
Pin 1	T31 (GND): Ground	Ground connection (-) for 12V voltage supply (Terminal 31)
Pin 2	Not connected	-
Pin 3	T87: 12V+ voltage supply	Battery connection (+) for 12V voltage supply; switched via relay (Terminal 87)
Pin 4	CAN-L	CAN low wire for CAN Bus
Pin 5	CAN-H	CAN high wire for CAN Bus
Pin 6	Not connected	-

<Pin assignment - communication plug away from belt pulley>



Pin	Connection	Description
Pin 1	Not connected	-
Pin 2	CAN-H	CAN high wire for CAN Bus
Pin 3	CAN-L	CAN low wire for CAN Bus
Pin 4	T87: 12V+ voltage supply	Battery connection (+) for 12V voltage supply; switched via relay (Terminal 87)
Pin 5	Not connected	-
Pin 6	T31 (GND): Ground	Ground connection (-) for 12V voltage supply (Terminal 31)

<Pin assignment - communication plug to belt pulley>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

4 Communication

Communication Protocol

The BRM uses CAN protocol 2.0B/Standard in Intel byte order. The used signals are listed and described in detail in this chapter.

E2E Protection

The End-to-End protection is disabled in this platform software since this is a customer specific feature.

Alive counter and checksum signals are sent and received but the received signals won't be processed or evaluated.

4.1 CAN-Matrix

The following table contains all the messages and signals including their definitions. These information are also part of the delivered CAN-Database (*.dbc file) and System Template (*.arxml file). Some signals are not supported in the current sample level. All signals are explained in the following chapters.

For better readability of this document in the following chapter most of the CAN signal names (written bold) are replaced with aliases (written italic).

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

direction BRM view	message	ID / cycle	signal	alias	unit	physical signal range [min/max]		bit witdh	value resolution	offset	default value
TX	Edrv_Act_1	0x2 / 10 ms	**Edrv_nrChksAct**	*Checksum actual torque*	-	0	255	8	1	0	-
			Edrv_nrAlvCntrAct	*Alive counter actual torque*	-	0	15	4	1	0	-
			Edrv_stRunAct	*Actual run state*	-	0	3	2	1	0	-
			Edrv_stOperModAct	*Actual operation mode*	-	0	15	4	1	0	-
			Edrv_tqAct	*Actual torque*	Nm	-200	209,5	12	0,1	-200	-
			Edrv_nAct	*Actual speed*	rpm	-24000	25146	13	6	-24000	-
			Edrv_uAct	*Actual DC voltage*	V	0	63,9375	10	0,0625	0	-
			Edrv_iAct	*Actual DC current*	A	-500	523	10	1	-500	-
TX	Edrv_Req_1	0x4 / 10 ms	**Edrv_nrChksReq**	*Checksum request*	-	0	255	8	1	0	-
			Edrv_nrAlvCntrReq	*Alive counter request*	-	0	15	4	1	0	-
			Edrv_tqReq	*Torque calculation EEM*	Nm	-200	209,5	12	0,1	-200	-
			Edrv_nReq	*Speed calculation EEM*	rpm	0	24570	12	6	0	-
			Edrv_stFltId	*Failure identification*	-	0	511	9	1	0	-
			Edrv_flgMilReqd	*Malfunction indicator lamp request*	-	0	1	1	1	0	-
			Edrv_ratTCritMac	*Machine temperature state*	%	0	126	6	2	0	-
			Edrv_ratTCritIvtr	*Inverter temperature state*	%	0	126	6	2	0	-
			Edrv_stTq	*Torque plausibilisation state*	-	0	2	2	1	0	-
			Edrv_stStrtPsbl	*Start possible state*	-	0	3	2	1	0	-
TX	Edrv_Pred_1	0x6 / 10 ms	**Edrv_nrChksLim**	*Checksum available torque*	-	0	255	8	1	0	-
			Edrv_nrAlvCntrLim	*Alive counter available torque*	-	0	15	4	1	0	-
			Edrv_tqMaxEemPrdn	*Maximum limited torque*	Nm	-200	209,5	12	0,1	-200	-
			Edrv_tqMaxAvl	*Maximum available torque*	Nm	-200	209,5	12	0,1	-200	-
			Edrv_tqMinEemPrdn	*Minimum limited torque*	Nm	-200	209,5	12	0,1	-200	-
			Edrv_tqMinAvl	*Minimum available torque*	Nm	-200	209,5	12	0,1	-200	-
			Edrv_stActDern	*Actual derating state*	-	0	15	4	1	0	-
TX	Edrv_Status_1	0x8 / 10 ms	**Edrv_sMask**	*Internal status mask*	-	-	-	64	-	-	-

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

direction BRM view	message	ID / cycle	signal	alias	unit	physical signal range [min/max]		bit witdh	value resolution	offset	default value
RX	Pt_Edrv_Des_1	0x1 / 10 ms	**Pt_nrChksAct**	*Checksum torque request*	-	0	255	8	1	0	0
			Pt_nrAlvCntrAct	*Alive counter torque request*	-	0	15	4	1	0	0
			Pt_stOperModDesEdrv	*Operation mode request*	-	0	15	4	1	0	0
			Pt_tqDesEdrv	*Torque request & Maximum allowed torque*	Nm	-200	209,5	12	0,1	-200	0
			Pt_tqMinEdrv	*Minimum allowed torque*	Nm	-200	209,5	12	0,1	-200	-50
			Pt_nDesEdrv	*Speed request*	rpm	-24000	25146	13	6	-24000	0
RX	Eem_Edrv_Pred_1	0x5 / 10 ms	**Eem_uMaxPredEdrv**	*Predicted maximum DC voltage*	V	10	73,75	8	0,25	10	54
			Eem_uMinPredEdrv	*Predicted minimum DC voltage*	V	0	63,75	8	0,25	0	54
			Eem_iMinPredEdrv	*Predicted minimum DC current*	A	-800	223	10	1	-800	-500
			Eem_iMaxPredEdrv	*Predicted maximum DC current*	A	-200	823	10	1	-200	500
RX	Eem_Edrv_Des_1	0xB / 10 ms	**Eem_nrChksAct**	*Checksum charging voltage request*	-	0	255	8	1	0	0
			Eem_nrAlvCntrAct	*Alive counter charging voltage request*	-	0	15	4	1	0	0
			Eem_uDesEdrv	*Charging voltage request*	V	0	63,9375	10	0,0625	0	48
			Eem_stRlyMvbEdrv	*48V-battery relay state*	-	0	3	2	1	0	0
RX	Eem_Edrv_Lim_1	0xD / 10 ms	**Eem_uMaxEdrv**	*Maximum allowed DC voltage*	V	0	63,9375	10	0,0625	0	52
			Eem_uMinEdrv	*Minimum allowed DC voltage*	V	0	63,9375	10	0,0625	0	36
			Eem_rMvbEdrv	*Estimated 48V-battery resistance*	Ohm	0	1,023	10	0,001	0	0,02
			Eem_iReqEdrv	*Current calculation EEM*	A	-500	523	10	1	-500	0
			Eem_iMaxEdrv	*Maximum allowed DC current*	A	0	1023	10	1	0	250
			Eem_iMinEdrv	*Minimum allowed DC current*	A	-1000	23	10	1	-1000	-250

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

direction BRM view	message	ID / cycle	signal	alias	unit	physical signal range [min/max]		bit witdh	value resolution	offset	default value
RX	Obd_State_1	0xE / 200 ms	Obd_flgCsersActv	*Cold start emission reduction strategy active*	-	0	1	1	1	0	0
			Obd_flgSrv04Enad	*Service 04 (clear diag info) enabled*	-	0	1	1	1	0	0
			Obd_stColdStrtDenomCdn	*Condition for evaporative IUMPR denominator*	-	0	2	2	1	0	0
			Obd_stDcycCdn	*Condition for confirmation of a pending fault code*	-	0	2	2	1	0	0
			Obd_stGnrlDenomCdn	*Condition for general IUMPR denominator*	-	0	2	2	1	0	0
			Obd_stGnrlDenomPdtcDltCdn	*Condition for PDTC delete*	-	0	1	1	1	0	0
			Obd_stIgnCycCdn	*Condition for IUMPR ignition cycle*	-	0	2	2	1	0	0
			Obd_stIgnCycEngRunngCdn	*Condition for ignition cycle*	-	0	2	2	1	0	0
			Obd_stWrmupCdn	*Condition for fault healing*	-	0	1	1	1	0	0
RX	Pt_Obd_1	0xF / 200 ms	Pt_dstOdo	*Odometer*	km	0	1,6E+06	24	0,1	0	0
			Pt_ratApp	*Accelerator pedal position*	%	0	100	8	0,5	0	0
			Pt_ratEngLoad	*Engine load*	%	0	100	8	0,5	0	0
			Pt_tCoolntEng	*Engine coolant temperature*	°C	-40	215	8	1	-40	120
			Pt_tiEngShOff	*Engine shut-off time*	min	0	4095	12	1	0	0
RX	Pt_Veh_1	0x10 / 100 ms	Pt_nrChksIgn	*Checksum ignition switch state*	-	0	255	8	1	0	0
			Pt_nrAlvCntrIgn	*Alive counter ignition switch state*	-	0	15	4	1	0	0
			Pt_stIgnSwt	*Ignition switch state*	-	0	7	3	1	0	4
			Pt_nEng	*Engine rotation speed*	rpm	0	16383	14	1	0	0
			Pt_vVeh	*Displayed vehicle speed*	km/h	0	409	12	0,1	0	0
			Pt_tAmbEdrv	*Ambient temperature E-Drive*	°C	-40	215	8	1	-40	120
			Pt_tiEdrvShOff	*E-Drive shut-off time*	min	0	4095	12	1	0	0
			Pt_flgCrash	*Crash flag*	-	0	1	1	1	0	0
RX	Pt_Obd_2	0x11 / 200 ms	Pt_tiSnceEngStrtd	*Time since engine started*	s	0	65535	16	1	0	0

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

4.2 Transmitted CAN signals by BRM

The signals are listed in the alphabetical order of their alias.

<Actual DC current>

Actual estimated DC link current of the BRM at B+ bolt.

Positive sign: current to BRM

Negative sign: current from BRM

<Actual DC voltage>

Actual measured BRM DC link voltage between B+ and B- bolt.

<Actual derating state>

Actual derating state of the BRM. Derating (power reduction) is active, if specific limit values (temperature, voltage or current) are reached.

0 = No derating

1 = Derating: DC voltage limitation

2 = Derating: DC current limitation

3 = Derating: torque limitation

4 = Derating: speed limitation

5 = Derating: inverter limitation

6 = Derating: motor temperature limitation

7 = Derating: phase current limit

<Actual operation mode>

Actual BRM operation mode

0x0 = Idle

0x1 = Torque mode

0x4 = Voltage mode

0x6 = Speed mode

0xB = Start mode

0xE = Standby ("Pre-Run", "Post-Run" or "Post-Run finished")

0xF = Failure

<Actual run state>

Actual BRM run state

0x0 = Pre-Run

0x1 = Run (Idle, Torque, ...)

0x2 = Post-Run

0x3 = Post-Run finished

<Actual speed>

Actual measured BRM speed.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

<Actual torque>

Actual estimated BRM torque at BRM shaft. The torque is calculated from internal measurement values of the BRM (speed, excitation current, phase currents). From BRM-perspective a positive value is a motor torque and a negative value is a generator torque.

<Alive counter actual torque>

Message counter of the related BRM message. The enumerator counts periodically and indicates that BRM microprocessor and the corresponding program is still running. The enumerator is calculated every 10 ms.

<Alive counter available torque>

Message counter of the BRM message Edrv_Pred_1. The enumerator counts periodically from 0 to 15 and indicates that BRM microprocessor and the corresponding program is still running. The enumerator is calculated every 10 ms.

<Alive counter request>

Message counter of the BRM message Edrv_Req_1. The enumerator counts periodically from 0 to 15 and indicates that BRM microprocessor and the corresponding program is still running. The enumerator is calculated every 10 ms.

<Checksum actual torque>

Checksum of the related BRM message. With the help of this checksum, receiving control units are able to verify the integrity of data in a message.

<Checksum available torque>

Checksum of the BRM message Edrv_Pred_1. With the help of this checksum, receiving control units are able to verify the integrity of data in a message.

<Checksum request>

Checksum of the BRM message Edrv_Req_1. With the help of this checksum, receiving control units are able to verify the integrity of data in a message.

<Failure identification>

The signal is not supported in this SW. The transmitted values are free of information.

<Internal status mask>

BRM internal status mask.

The signal is not supported in this SW. The transmitted values are free of information.

<Inverter temperature state>

Actual ratio of the BRM Inverter temperature in comparison to the maximum permissible Inverter temperature. Values between 75 % and 100 % lead to a torque derating to prevent the BRM from overheating.

<Machine temperature state>

Actual ratio of the BRM electric machine temperature in comparison to the maximum permissible machine temperature. Values between 75 % and 100 % lead to a torque derating to prevent the BRM from overheating.

<Malfunction indicator lamp request>

BRM request towards the MIL master to activate the MIL (Malfunction Indication Lamp).

The signal is not supported in this SW. The transmitted values are free of information.

<Maximum available torque>

Calculated maximum BRM torque based on peak performance and thermal derating of BRM.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

<Maximum limited torque>

Calculated maximum BRM torque based on peak performance, thermal derating, electrical derating and EEM electric prediction values <predicted minimum DC voltage> & <predicted maximum DC current>.

<Minimum available torque>

Calculated minimum BRM torque based on peak performance and thermal derating of BRM.

<Minimum limited torque>

Calculated minimum BRM torque based on peak performance, thermal derating, electrical derating and EEM electric prediction values <predicted maximum DC voltage> & <predicted minimum DC current>.

<Speed calculation EEM>

Calculated BRM speed based on EEM electrical current request value <current calculation EEM>.

See also <Torque calculation EEM>.

<Start possible state>

Indicates that a start of the combustion engine can be performed in terms of BRM thermal ratio. Electrical limits (i.e. current or voltage) are not considered.

0x0 = Start Possible

0x1 = Start Critical

0x2 = Start Not Possible

<Torque calculation EEM>

Calculated BRM torque based on EEM electrical current request value <current calculation EEM>.

See also <Speed calculation EEM>

<Torque plausibilisation state>

0 = Torque is plausible

1 = Torque is bigger

2 = Torque is smaller

3 = Torque signal not available

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

4.3 Received CAN Signals by BRM

<48V-battery relay state>

Actual state of 48V (Medium Voltage) Battery main relay.

0 = Closed

1 = Opening request

2 = Open

3 = Precharge

The implementation of this signal is not complete yet. State 0 ("Closed") shall be sent always.

<Accelerator pedal position>

This signal will not be evaluated in this SW. The received values will not be processed.

<Alive counter charging voltage request>

Message counter of the EEM message Eem_Edrv_Des_1. The counter runs cyclically from 0 to 15 and indicates if the EEM microprocessor and the corresponding program are still running.

The signal will not be evaluated in this SW. The received values will not be processed.

<Alive counter ignition switch state>

Message counter of the PT message Pt_Veh_1. The counter runs cyclically from 0 to 15 and indicates if the PT microprocessor and the corresponding program are still running.

This signal will not be evaluated in this SW. The received values will not be processed.

<Alive counter torque request>

Message counter of the related message. The counter runs cyclically and indicates if the PT microprocessor and the corresponding program are still running.

This signal will not be evaluated in this SW. The received values will not be processed.

<Ambient temperature E-Drive>

This signal will not be evaluated in this SW. The received values will not be processed.

<Charging voltage request>

Charging DC voltage request for the BRM. This is the set point value for the BRM DC voltage, in case the <operation mode request> is "voltage mode".

<Checksum charging voltage request>

Checksum of EEM message Eem_Edrv_Des_1. With the help of this checksum, receiving control units are able to verify the integrity of data in a message.

The signal will not be evaluated in this SW. The received values will not be processed.

<Checksum ignition switch state>

Checksum of PT message Pt_Veh_1. With the help of this checksum, receiving control units are able to verify the integrity of data in a message.

This signal will not be evaluated in this SW. The received values will not be processed.

<Checksum torque request>

Checksum of the related message. With the help of this checksum, receiving control units are able to verify the integrity of data in a message.

This signal will not be evaluated in this SW. The received values will not be processed.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

<Cold start emission reduction strategy active>

This signal will not be evaluated in this SW. The received values will not be processed.

<Condition for confirmation of a pending fault code>

This signal will not be evaluated in this SW. The received values will not be processed.

<Condition for evaporative IUMPR denominator>

This signal will not be evaluated in this SW. The received values will not be processed.

<Condition for fault healing>

This signal will not be evaluated in this SW. The received values will not be processed.

<Condition for general IUMPR denominator>

This signal will not be evaluated in this SW. The received values will not be processed.

<Condition for ignition cycle>

This signal will not be evaluated in this SW. The received values will not be processed.

<Condition for IUMPR ignition cycle>

This signal will not be evaluated in this SW. The received values will not be processed.

<Condition for PDTC delete>

This signal will not be evaluated in this SW. The received values will not be processed.

<Crash flag>

Crash indication to BRM

<Current calculation EEM>

Requested electrical power from BRM (in combination with actual BRM voltage). With the help of this value, the BRM calculates a desired mechanical power and provides it in the signals <torque calculation EEM> and <speed calculation EEM>.

<Displayed vehicle speed>

This signal will not be evaluated in this SW. The received values will not be processed.

<E-Drive shut-off time>

Time from BRM shutdown to BRM startup. This time influences OBD plausibilisation of the temperature sensor and thermal derating because it is used to calibrate the temperature model.

<Engine coolant temperature>

This signal will not be evaluated in this SW. The received values will not be processed.

<Engine load>

This signal will not be evaluated in this SW. The received values will not be processed.

<Engine rotation speed>

This signal will not be evaluated in this SW. The received values will not be processed.

<Engine shut-off time>

This signal will not be evaluated in this SW. The received values will not be processed.

<Estimated 48V-battery resistance>

Actual 48V (Medium Voltage) Battery internal resistance. This value is used for BRM voltage limitation.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

<Ignition switch state>

Ignition switch. This signal is used to control BRM run state:

0x0 = IGN_LOCK ("off")

0x1 = IGN_OFF ("off")

0x2 = IGN_ACC ("on")

0x4 = IGN_ON ("on")

0x5 = IGN_START ("on")

<Maximum allowed DC current>

Maximum allowed DC current at B+ bolt of the BRM. This limit will be considered as positive current in all active operation modes.

<Maximum allowed DC voltage>

Maximum allowed DC voltage between B+ and B- bolt of the BRM. This limit will be considered as maximum voltage in the following operation modes: torque mode, speed mode and start mode.

<Maximum allowed torque>

Maximum allowed BRM torque at BRM shaft. This value is considered as maximum torque value in all active operation modes.

<Minimum allowed DC current>

Minimum allowed DC current at B+ bolt of the BRM. This limit will be considered as negative current in all active operation modes.

<Minimum allowed DC voltage>

Minimum allowed DC voltage between B+ and B- bolt of the BRM. This limit will be considered as minimum voltage in the following operation modes: torque mode, speed mode and start mode.

<Minimum allowed torque>

Minimum allowed BRM torque at BRM shaft (maximum generator torque). This value is considered as minimum allowed torque limitation in all active operation modes.

<Odometer>

This signal will not be evaluated in this SW. The received values will not be processed.

<Operation mode request>

Desired BRM operation mode:

0x0 = Idle

0x1 = Torque mode

0x4 = Voltage mode

0x6 = Speed mode

0xB = Start mode

<Predicted maximum DC current>

Predicted maximum DC current (maximum motor current) by EEM. The signal is used in combination with <predicted minimum DC voltage> to indicate a predicted maximum electric power for BRM. The predicted electric power serves as the basis for calculation of <maximum limited torque>.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

<Predicted maximum DC voltage>

Predicted maximum DC voltage by EEM. The signal is used in combination with <predicted minimum DC current> to indicate a predicted minimum electric power for BRM. The predicted electric power serves as the basis for calculation of <minimum limited torque>.

<Predicted minimum DC current>

Predicted minimum DC current (maximum generator current) by EEM. The signal is used in combination with <predicted maximum DC voltage> to indicate a predicted minimum electric power for BRM. The predicted electric power serves as the basis for calculation of <minimum limited torque>.

<Predicted minimum DC voltage>

Predicted minimum DC voltage by EEM. The signal is used in combination with <predicted maximum DC current> to indicate a predicted maximum electric power for BRM. The predicted electric power serves as the basis for calculation of <maximum limited torque>.

<Service 04 (clear diag info) requested>

This signal will not be evaluated in this SW. The received values will not be processed.

<Speed request>

Desired BRM speed at BRM shaft. Desired speed is considered as target value in speed mode.

<Time since engine started>

Time since engine started.

<Torque request>

Requested BRM torque at BRM shaft. The torque request is considered as target value in torque mode.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

5 BRM Initialization

Due to the complexity of the E-Machine and its functionalities, this chapter provides easy step-by-step instructions for the first initiation of the BRM. For this initiation it does not matter whether the BRM will be implemented in a prototype vehicle or integrated in a specific (functional) test bench.

The following steps are chronically ordered. Interchanging the order may lead to a malfunction of the BRM or to fault memory entries. For this reason it is recommended to perform the initiation in the order shown here.

Mechanical connection:

The BRM has to be mounted torsion-proof. The 48 V plus- and ground wires have to be bolted shock and vibration resistant to the B+ and B- bolt. The signal plug has to be wired and connected according to chapter <Communication-Interface>.

Important: The ground connections 12 V and 48 V have to be based on the same electrical potential to ensure a faultless function.

Communication interface:

The CAN signals for controlling the BRM can be generated with the provided CANoe configuration. Preconfigured signals are provided in the "Interactive Generator" to provide the required BRM input signals.

Important: The preconfigured signal minimum/maximum values have to be adapted:

- voltage and current limits according to the supply voltage or 48V battery limits.
- the default signal values from chapter <CAN-Matrix> can be used for the other signals.

Electrical connection:

After starting the CANoe configuration, the 12 V can be applied at the signal plug.

After the voltage is applied, the BRM ECU will start working. This will become obvious (after starting resp. pushing F9) through the appearance of the BRM TX messages in the Trace window of the CANoe configuration.

From now on, the signal generator shall send the <operation mode request> = "Idle" and <ignition switch state> = "on". Afterwards the 48 V can be applied at the B+ bolt by using an active current limitation (max 100 A!).

Important: Powering up the 48 V without current limitation may lead to a destruction of the DC link voltage capacitors within the inverter.

Control:

If the value <ignition switch state> = "off", the BRM will switch automatically into <actual run state> = "Post-Run finished" in case the 48 V voltage has not reached operating level within 10 s. If this happens, a reset can be commanded by <ignition switch state> = "on".

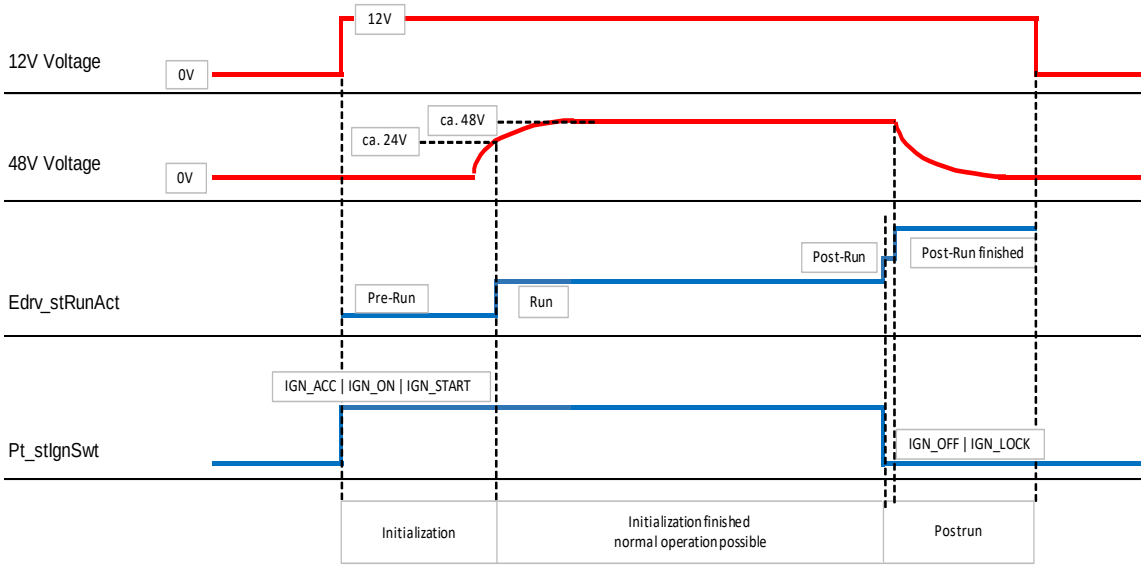
If the pre-run Tests are passed successfully, BRM will enter the run state and send <actual run state> = "Run" and <actual operation mode> = "Idle". With the signal generator, it is possible to send a <torque request> value (e.g. - 5 Nm for a generator torque). As soon as <operation mode request> is set to "Torque mode", the BRM returns <actual operation mode> = "Torque mode" and provides the requested torque.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

The value "off" shall be sent in the <ignition switch state> via the signal generator before disconnecting the 12 V and 48 V voltage. During the Post-Run procedure, the BRM sends at first <actual run state> = "Post-Run" followed by <actual run state> = "Post-Run finished". From now on the 12 V and 48 V voltage may be disconnected. Alternatively a restart without switching off/on voltages can be commanded by <ignition switch state> = "on". Once the 48V voltage level has dropped close to 0V, the 12V supply voltage can be switched off.

Note: After the 48V side has been charged, there may occur situations when this Voltage level remains high on BRM B+/B- even if the BRM is no longer connected to any battery or supply. If this shall be avoided, the vehicle system has to grant an external active discharge.

The procedure for initialization and deactivation is shown in Image <Initialization and Shut Down>.



<Initialization and Shut Down>

Measurement:

The CANoe configuration can also be used to measure the signal flow BRM ←→ System.

In the Tab "View“, one can find the window "Simulation Setup“. The measurement can be started by activating the lowest block "Logging“. The path for saving can be set by clicking on the folder.

By double clicking on the switch at the left side, the saved measurement values can be read as offline values and can be presented in multiple ways.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

6 Operation Modes

The control strategy is coordinated by the superior control units. The requested operation mode is communicated with the signal <operation mode request> to the BRM. The possible operation modes are described in the following section.

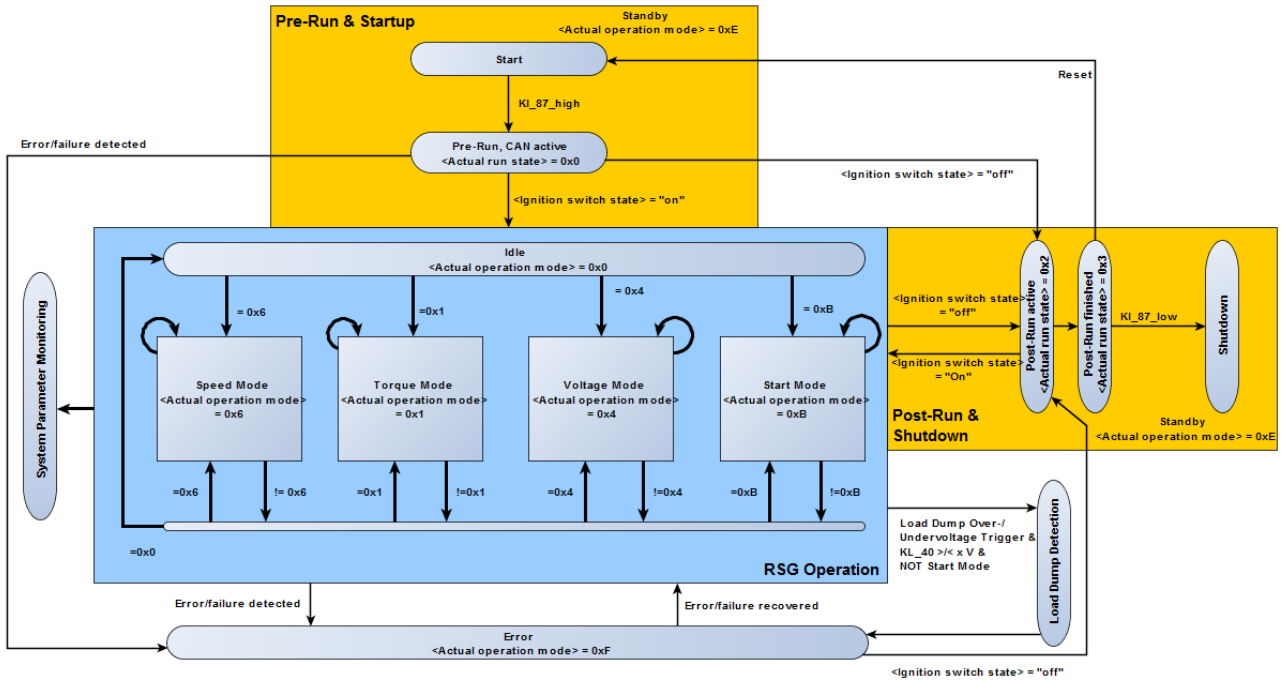
6.1 State Machine

BRM is able to start up (Transition from "Pre-Run" to "Idle"/"Run"), although the BRM already rotates before "Pre-Run" is finished.

All active operation modes are only available in "Run" state.

It is possible to change directly between the operation modes torque mode, speed mode and voltage mode.

Note that Voltage Mode can only be entered if rotating speed > 600rpm on BRM side. To get a negative torque, the BRM speed should be > 1200rpm, so that the voltage level can actually be controlled in a robust way.



<State Machine>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

6.2 Torque Mode

The Torque mode can be used for several operation strategy use cases. For example Boost or Recuperation. In torque mode the BRM can operate in motor mode as well as in generator mode.

The Torque mode is selected via <operation mode request> = "Torque mode". BRM responds <actual operation mode> = "Torque mode" in torque mode. If the requested torque is zero (<Torque request> = 0 Nm), the BRM will automatically switch to "Idle" after 0.5 s to reduce losses (in case rotation speed on BRM side > 885 rpm).

The superior control unit is responsible for the boost and recuperation strategy. The requested operation mode and the requested torque can be communicated to the BRM via the signals <operation mode request> and <torque request>. The change in the operation mode will be acknowledged by the BRM in the signal <actual operation mode>. The actual torque is returned back from the BRM in the signal <actual torque>.

Important:

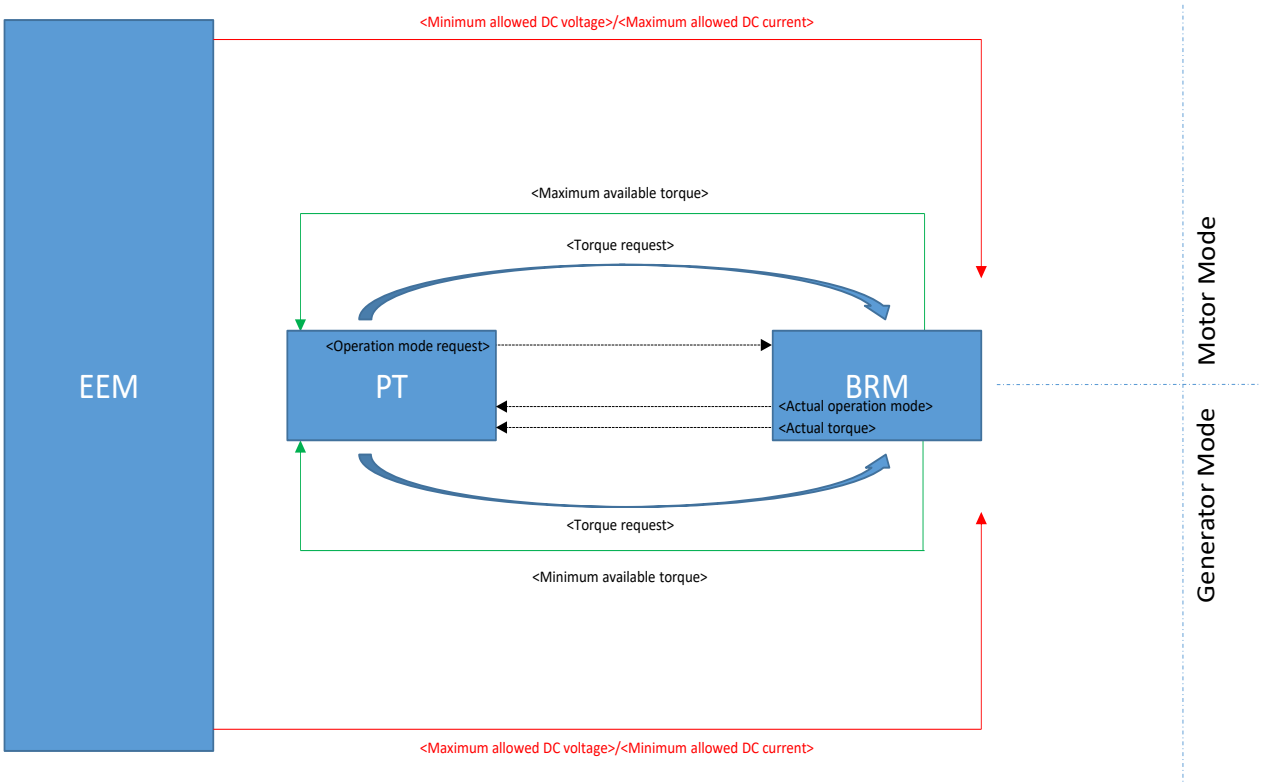
- The values in the signal <torque request> shall not be smaller than in the signal <minimum allowed torque>. By default the same values shall be sent in both signals.
- Do not send the BRM output signal <maximum limited torque> back to the BRM as input signals <torque request> and <maximum allowed torque> (both are mapped to the same can signal). If the signal is directly fed back or limited to this signal, the torque request will be limited to 0 Nm and no BRM operation will be possible in torque mode. Please use the signal <maximum available torque> instead of <maximum limited torque>.

For the purpose of electric power limitation, the superior control unit sends specific values for current and voltage limitation to the BRM, which will be considered (motor mode: <minimum allowed DC voltage> & <maximum allowed DC current>; generator mode: <maximum allowed DC voltage> & <minimum allowed DC current>).

The minimum/maximum achievable BRM torque with respect to temperature limitation will be continuously estimated and sent in the signals <minimum available torque> and <maximum available torque>.

The signal flow is displayed in the Image below.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Control Strategy - Torque Mode>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

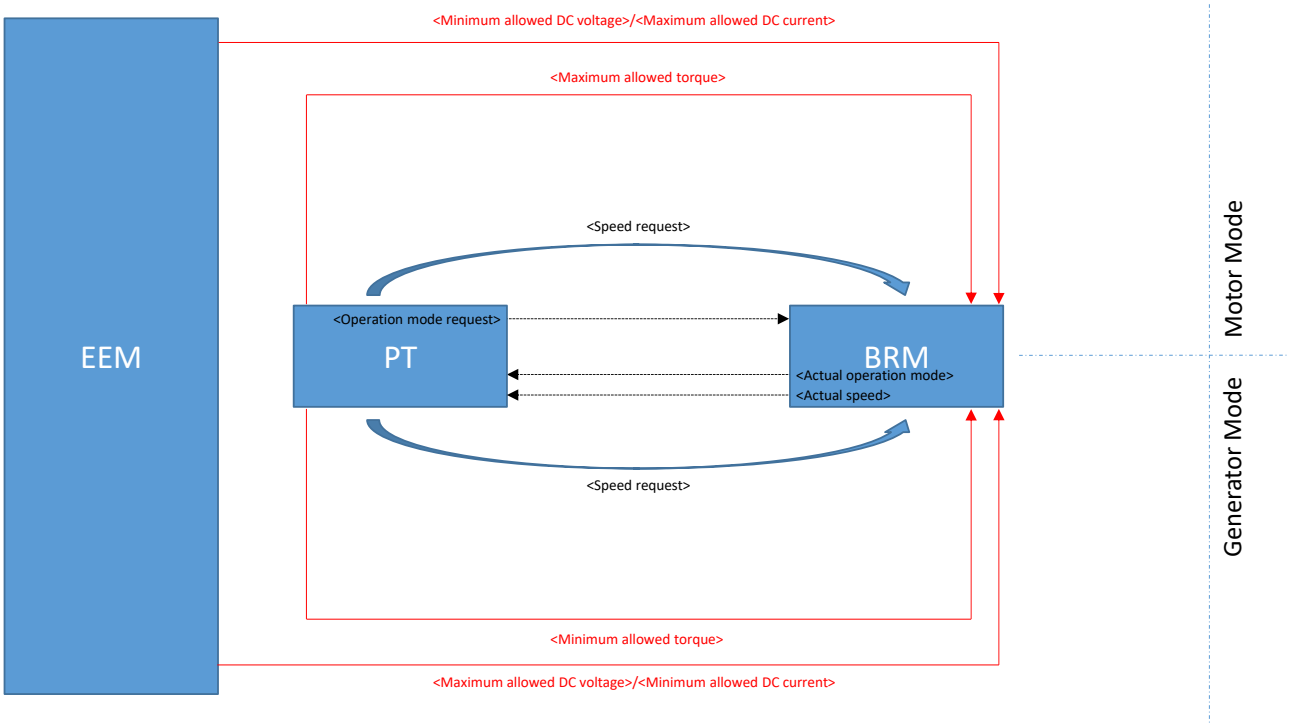
6.3 Speed mode

The Speed mode enables a BRM operation with a closed loop speed control (by using a closed loop torque control) with a rotation speed target. This allows i.e. the realization of motor and generator operation strategies such as engine start or holding a constant speed after engine start.

The superior control unit sends the signal <operation mode request> = "Speed mode" and a requested rotation speed in the signal <speed request>. The BRM acknowledges a change in the operation mode in the signal <actual operation mode>. The actual machine speed will be returned in the signal <actual speed>. The torque available values and torque limited values are still available.

For the purpose of electric power limitation, the BRM receives specific values for DC current and DC voltage limitation, which will be considered in either case (motor mode: <minimum allowed DC voltage> & <maximum allowed DC current>; generator mode: <maximum allowed DC voltage> & <minimum allowed DC current>).

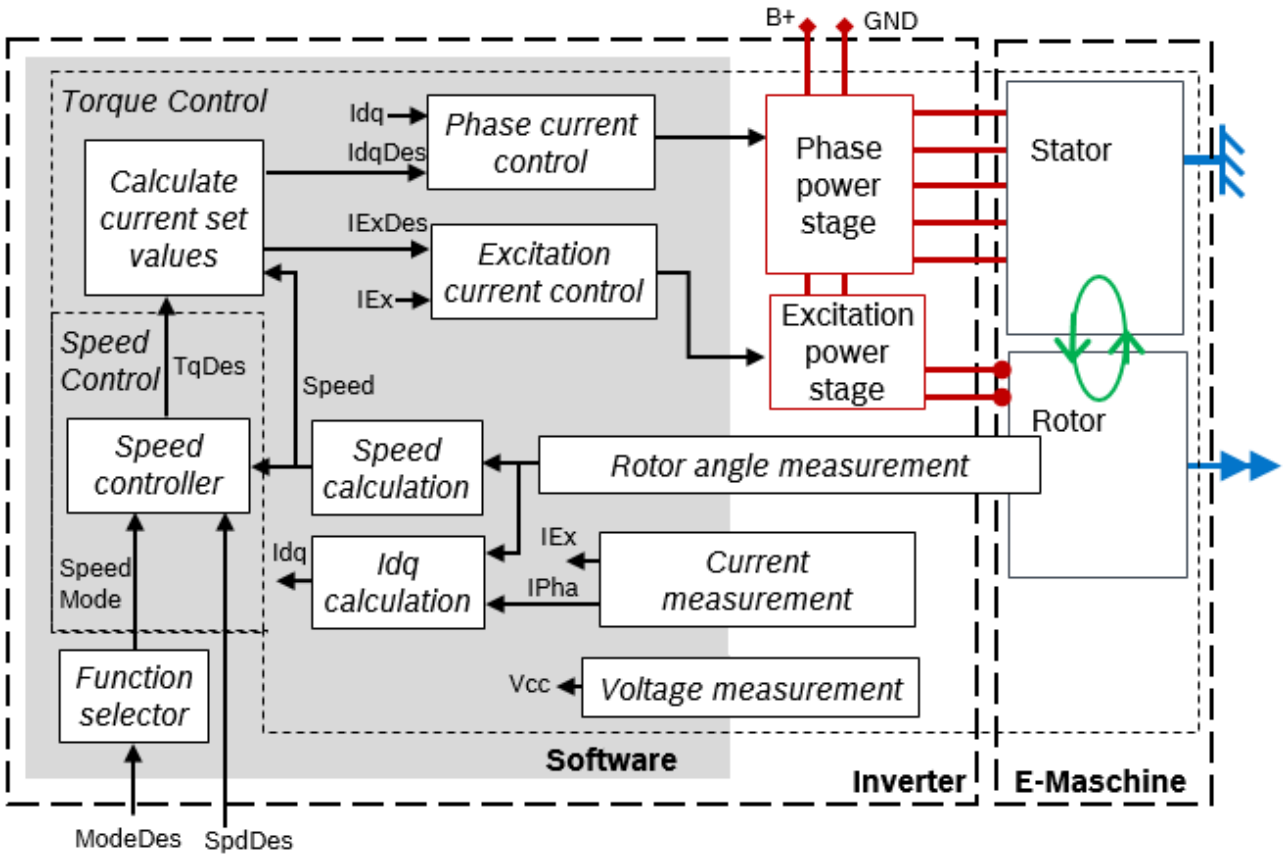
For the purpose of torque limitation, the BRM receives <maximum allowed torque> and <minimum allowed torque>, which will be considered by the BRM in speed mode.



<Control Strategy - Speed Mode>

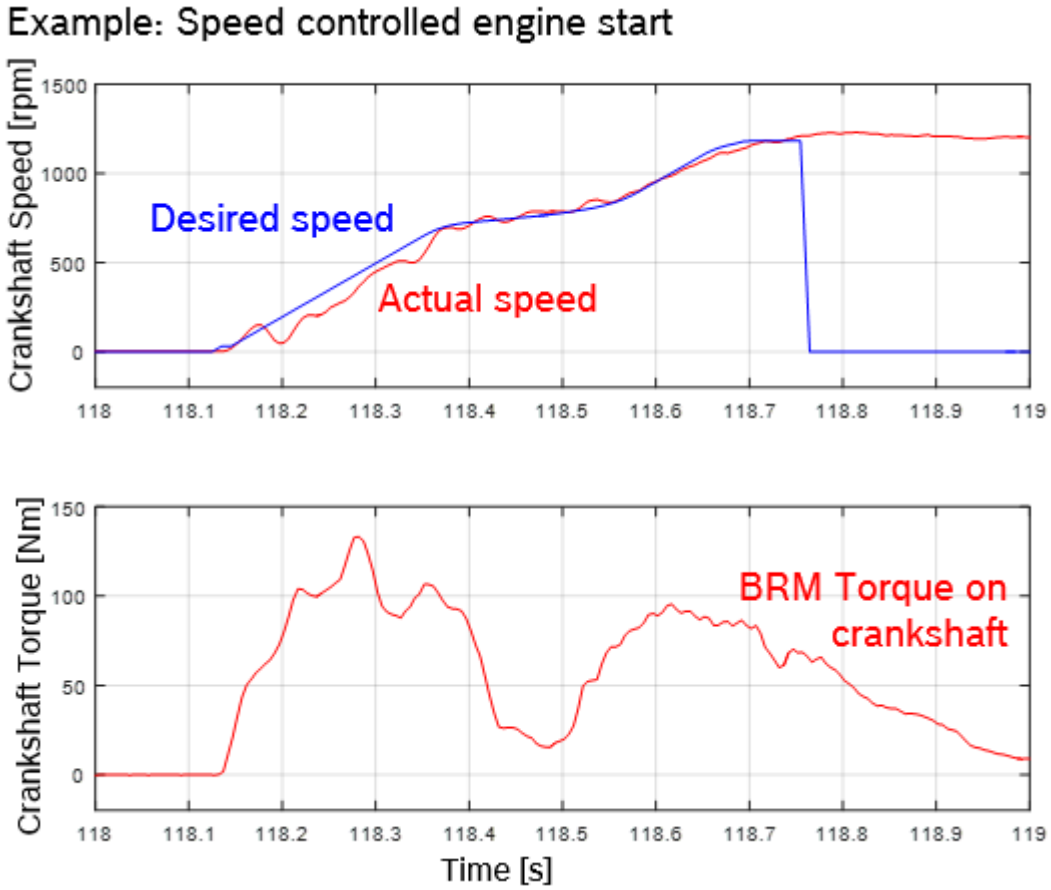
The image <Internal function schematics - Speed control> gives an overview about the internal implementation of the speed controller in the BRM SW. The result achieved in a typical application of the speed controller is shown in image <Example of speed controlled engine start> (please note that the BRM speed and torque are displayed on crankshaft level in this example).

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Internal function schematics - Speed control>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Example of speed controlled engine start>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

6.4 Voltage mode

The Voltage mode enables a BRM operation with a closed loop voltage control with a DC B+ link voltage target for BRM.

To ensure that the engine is never stalled and to be able to control the voltage level in a robust way,

- 600 RPM (at BRM) is the minimum rotation speed for entering voltage control mode
- Between 600 RPM (at BRM) and 1200 RPM only partial requests are processed (to positive torque)
- Between 1200 RPM (at BRM) and 1500 RPM (at BRM) generator requests are "ramped in" (and processed to negative torque)
- From 1500 RPM on motor and generator requests are fully processed.

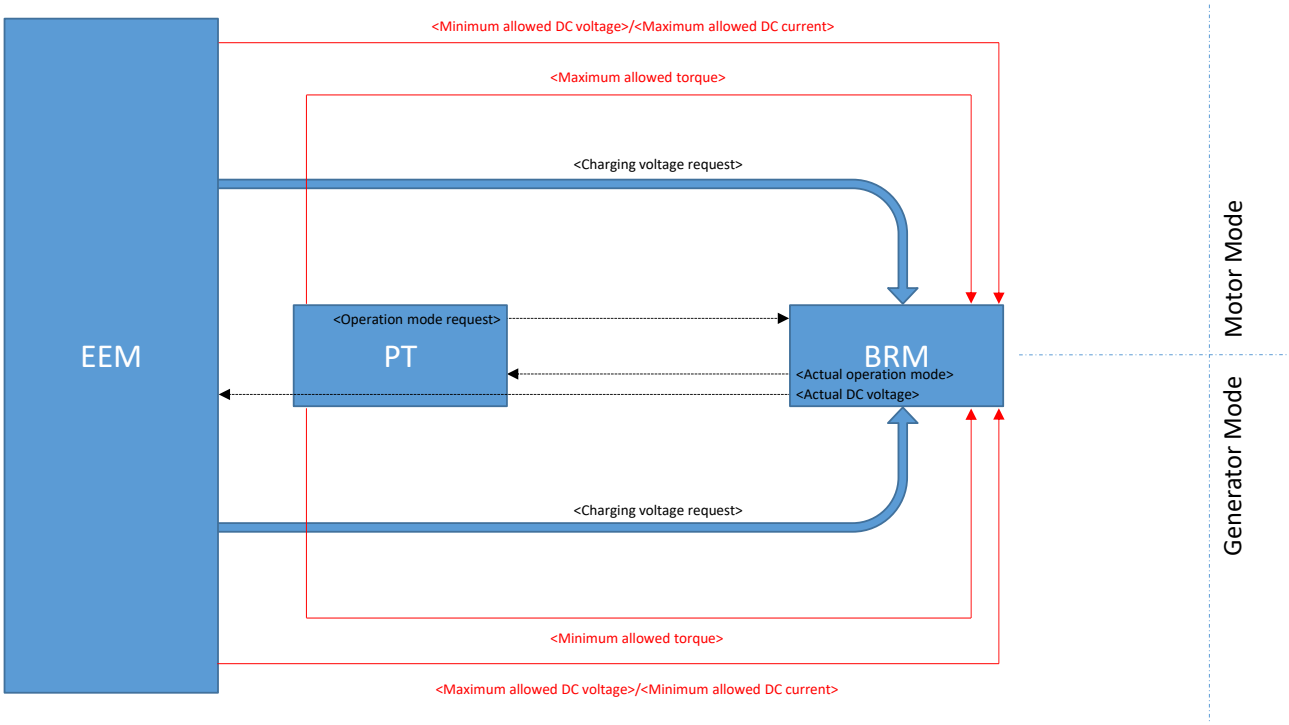
The superior control unit provides the target value for the voltage mode within the signal <charging voltage request>. The requested operation mode shall be set in the signal <operation mode request> = "Voltage mode" to the BRM. The BRM acknowledges the change in the signal <actual operation mode> = "Voltage mode" and returns the measured actual voltage in the signal <actual DC voltage>. Motor and generator operation is possible in this mode.

For limitation of BRM torque, the superior control unit sends the signals <maximum allowed torque> and <minimum allowed torque> to the BRM. The voltage limits within the signals <maximum allowed DC voltage> and <minimum allowed DC voltage> will not be taken into account within this operation mode. The DC current limitation of the BRM is still considering the input signals <maximum allowed DC current> and <minimum allowed DC current>.

The torque prediction values and torque derating values are still available.

The signal flow is displayed in the Image below.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Control Strategy - Voltage Mode>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

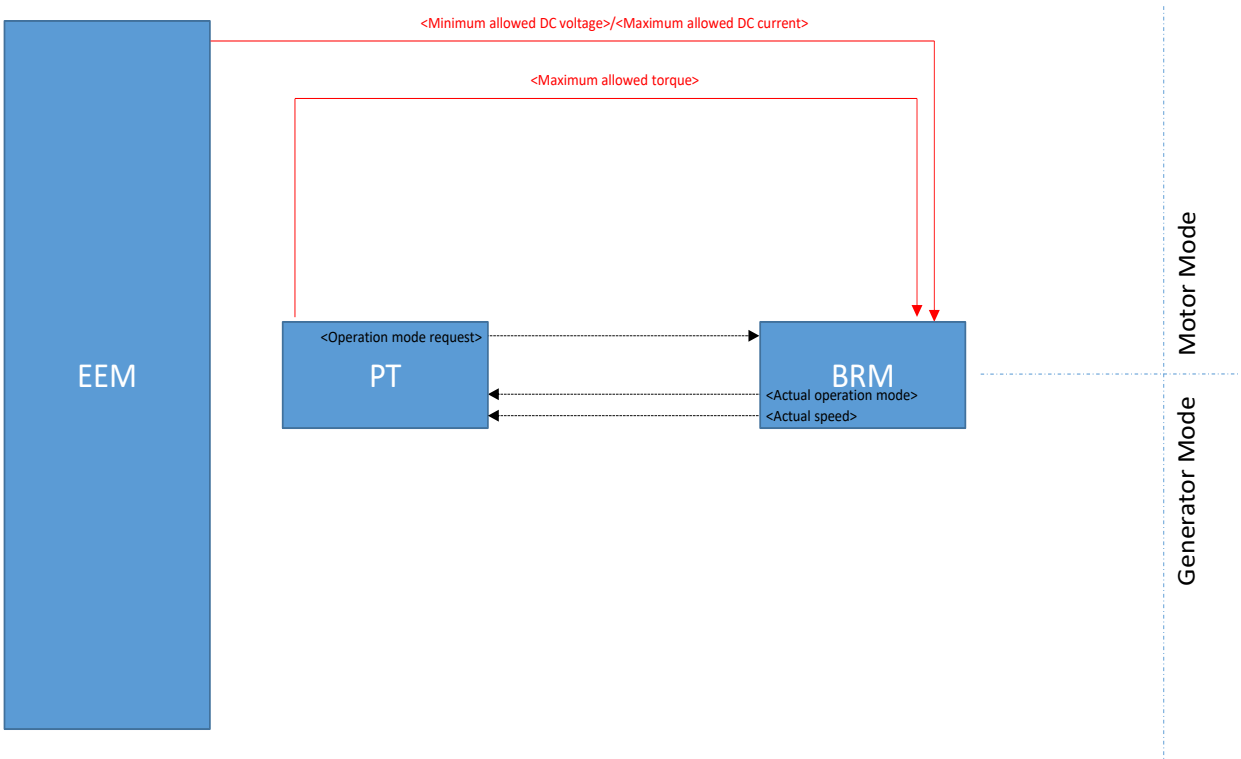
6.5 Start mode

The superior control unit provides the target values for the Start mode. The signal <operation mode request> = "Start mode" activates the Start mode. The BRM acknowledges a change in the operation mode in the signal <actual operation mode> = "Start mode".

The superior control units sends a maximum torque in the signal <maximum allowed torque> (e.g. 40 Nm) and subsequently changes the operation mode <operation mode request> = "Start mode". In case of a motor start, the BRM follows an internal torque-speed curve. The interval torque-speed curve is displayed in the image <Internal torque request versus rotation speed in Start mode>. The internal torque is limited to the signal <maximum allowed torque>.

For the purpose of electric power limitation, the superior control unit sends specific values for DC current and DC voltage limitation to the BRM, which will be considered (<minimum allowed DC voltage> & <maximum allowed DC current>).

The signal flow is displayed in the Image below.

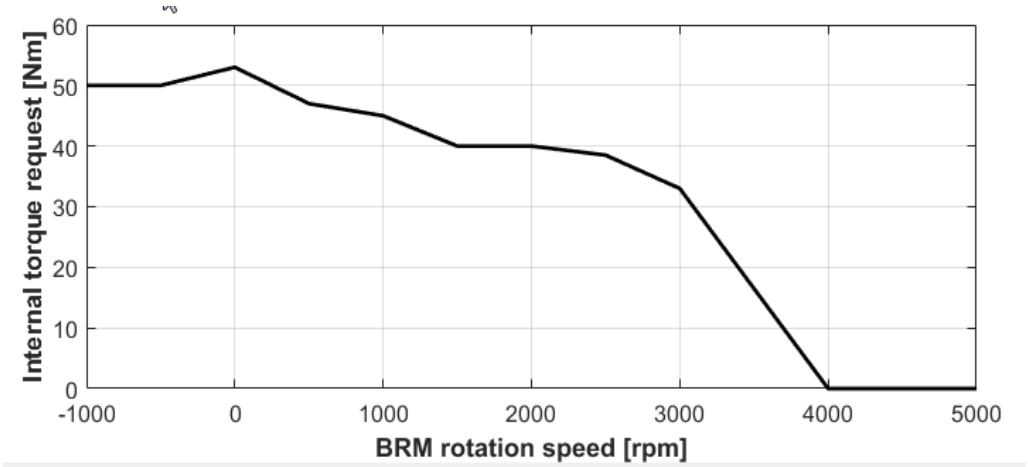


<Control Strategy - Start Mode>

The BRM stays in "start mode" and provides torque depending on the implemented torque curve (see <internal torque request versus rotation speed in Start mode>).

For BRM-rotation-speed > 4000 rpm the provided torque is 0 Nm.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Internal torque request versus rotation speed in Start mode>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

6.6 BRM Concept for Derating, Limitation & Prediction

The BRM includes several of functions for the purposes of thermal derating, torque limitation & torque prediction. The following aspects have an influence on the BRM torque:

- General torque limitation

  Depending on actual voltage and actual speed the actual BRM peak torque is determined. This minimum/maximum torque is only depending on BRM internal values. At rated conditions of speed and voltage this minimum/maximum torque matches the peak torque values described in the TCD/TKU.

- Thermal derating

  Please refer to detailed description below.

- Electric limitation (voltage & current)

  Please refer to detailed description below.

The following aspects have no influence on the BRM torque:

- External prediction (power based)

  For some applications it is useful to provide estimated torque prediction limits in order to improve system behavior. This prediction is intended to be used as an information without having any influence on the electric limitation. This minimum/maximum torque is only depending on external values that are provided by CAN signals (EEM prediction signals). The estimation of torque prediction limits is done with a power based calculation approach: The provided voltage and current values are multiplied to an electric power. With an estimated efficiency and the actual BRM speed, this electric power will be converted to a torque value. For generator operation a constant efficiency of 83 % is used for calculation. For motor operation a constant efficiency of 82 % is used for calculation. The change of BRM speed or BRM temperature is not predicted. Since a power based approach is used the output values should also not be used in case of zero speed (no calculation of torque from mechanical power possible).

Implementation of thermal derating

If any thermal component of the BRM is reaching its temperature limit, the minimum/maximum available torque gets reduced continuously. The thermal components of the BRM are stator, rotor, MOSFETs, logic and choke. This minimum/maximum available torque is only depending on BRM internal values (e.g. estimated & measured temperatures).

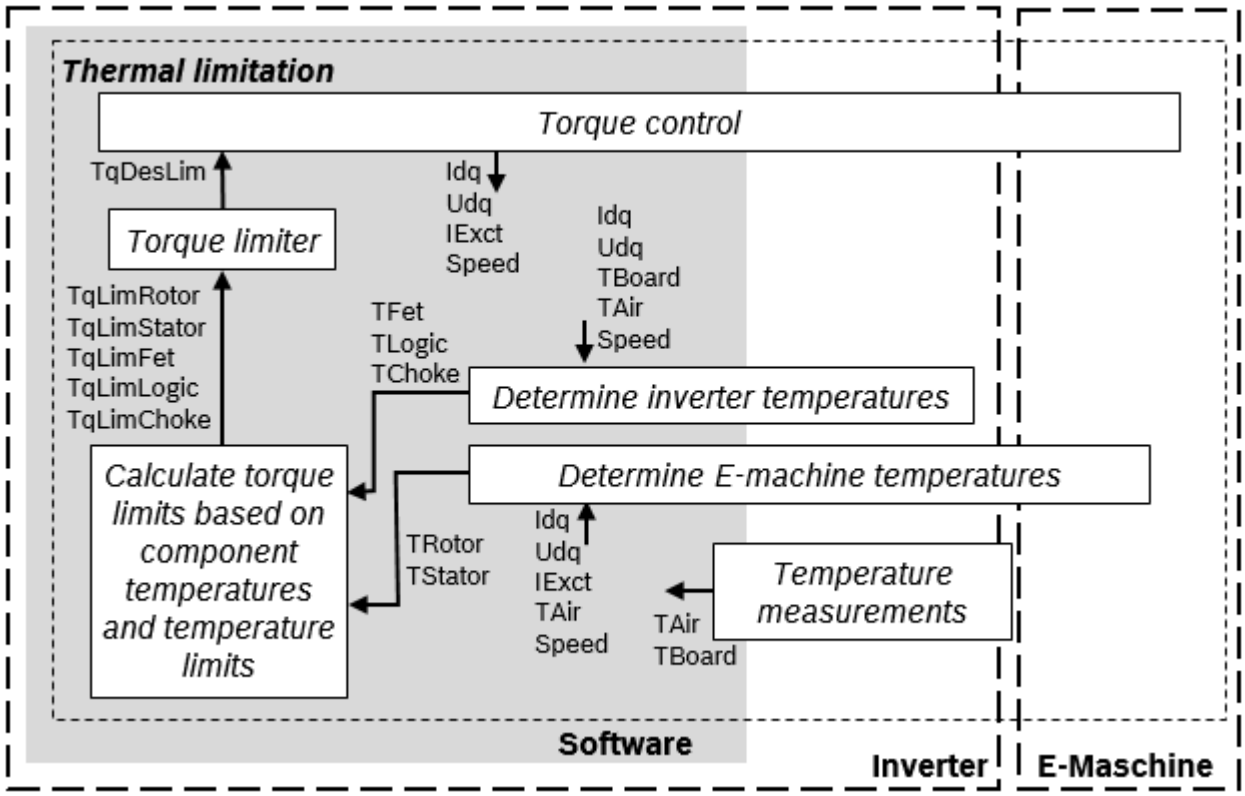
The overall thermal situation of the BRM is provided by the CAN signals <Machine temperature state> and <Inverter temperature state>.

For verification of internal thermal model temperatures an injection function is implemented. This function estimates the stator temperature through the measurement of the electrical resistance of the phases. The injection is performed initially once the engine is started. A cyclic repetition of the function starts from a stator temperature of 180° Celsius. The injection function can be detected through a short current pulse in

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

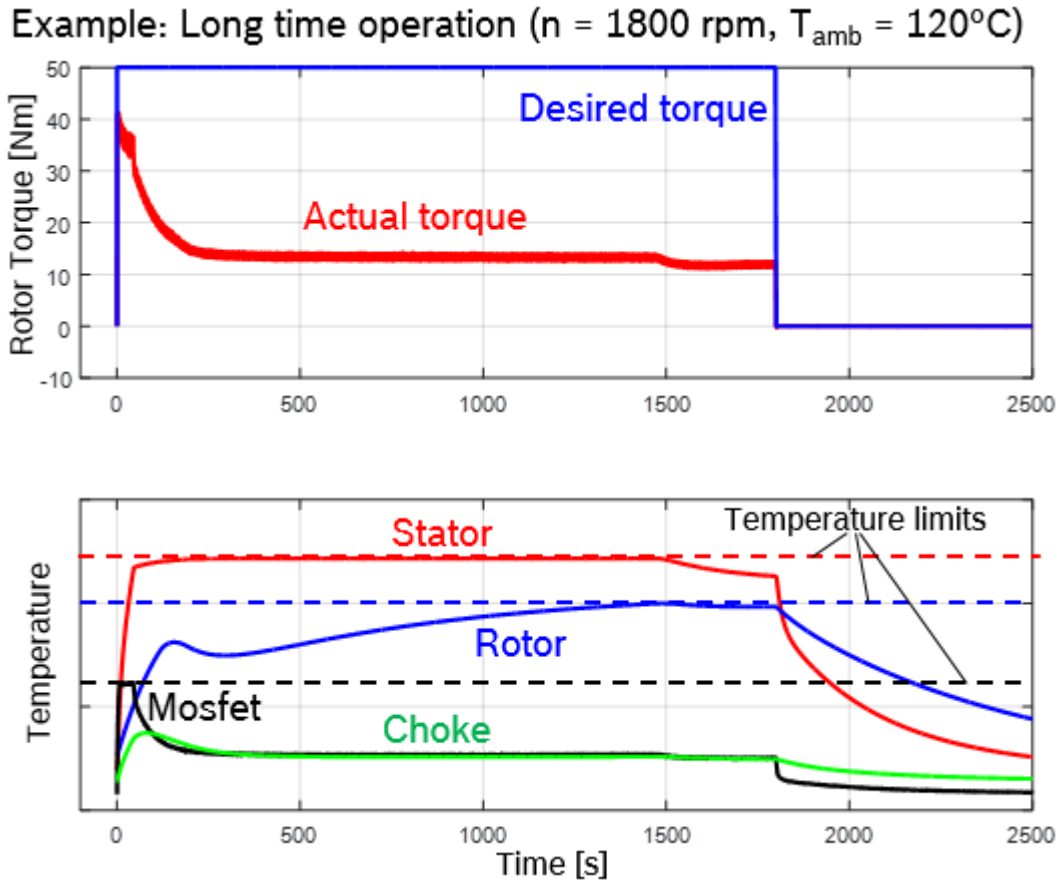
idle mode for 0.5 seconds with a small torque output (<Actual torque> is appr. -1Nm dependend on the <Actual speed>).

The image <Internal function schematics - Thermal derating> gives an overview about the internal implementation of the thermal derating function in the BRM SW. A possible result achieved with this thermal derating function is shown in image <Example of active thermal derating during long term operation>.



<Internal function schematics - Thermal derating>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Example of active thermal derating during long term operation>

Implementation of electrical limitation

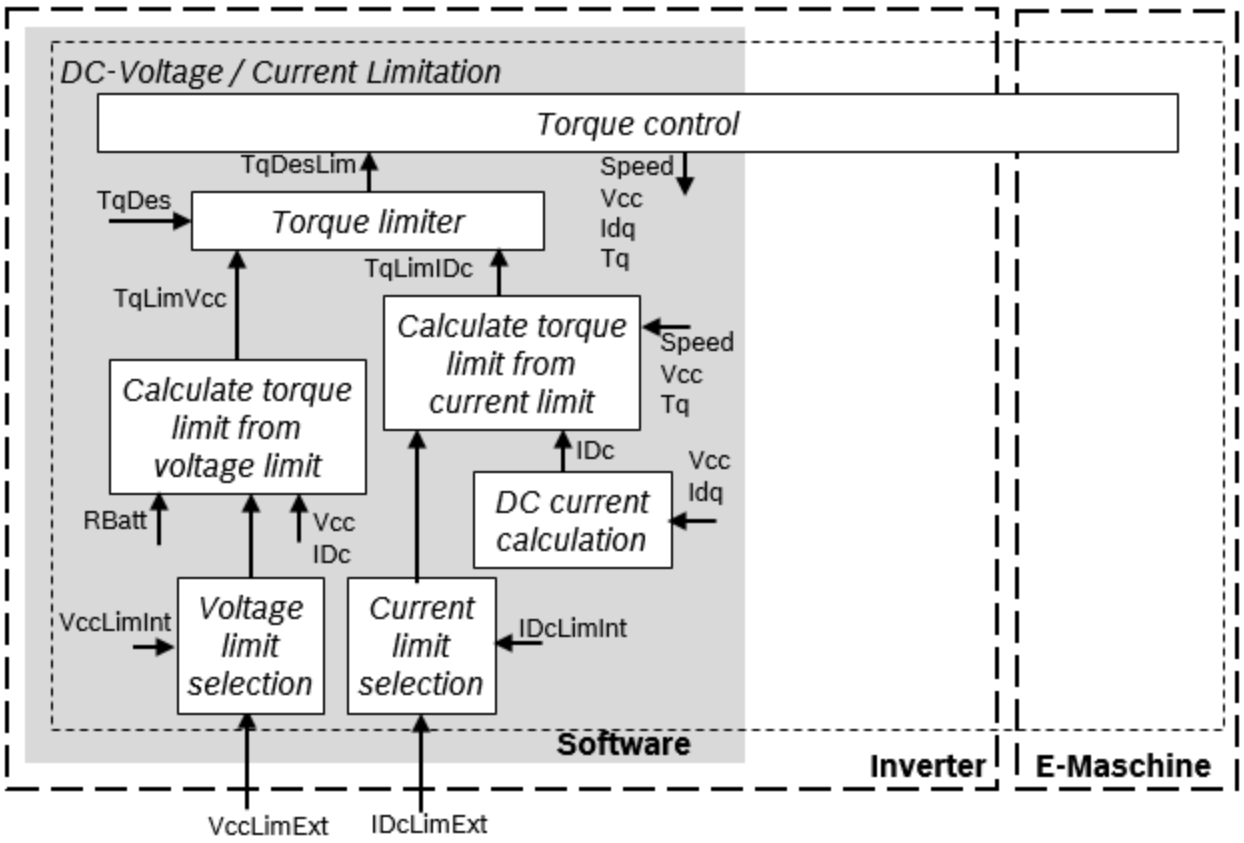
If the actual BRM voltage or current is reaching the voltage limit or current limit, the minimum/maximum limited torque gets reduced continuously. This minimum/maximum limited torque is depending on BRM internal values (i.e. actual voltage, actual current, internal voltage & current limits) as well as on external values. These external values are provided with the CAN signals for voltage & current limitation:

- <Maximum allowed DC voltage>
- <Minimum allowed DC voltage>
- <Maximum allowed DC current>
- <Minimum allowed DC current>

In order to enable a good accuracy of the voltage limitation over a wide range of boundary conditions, the actual estimated 48V battery resistance shall be provided to the BRM in the can signal <Estimated 48V-battery resistance>.

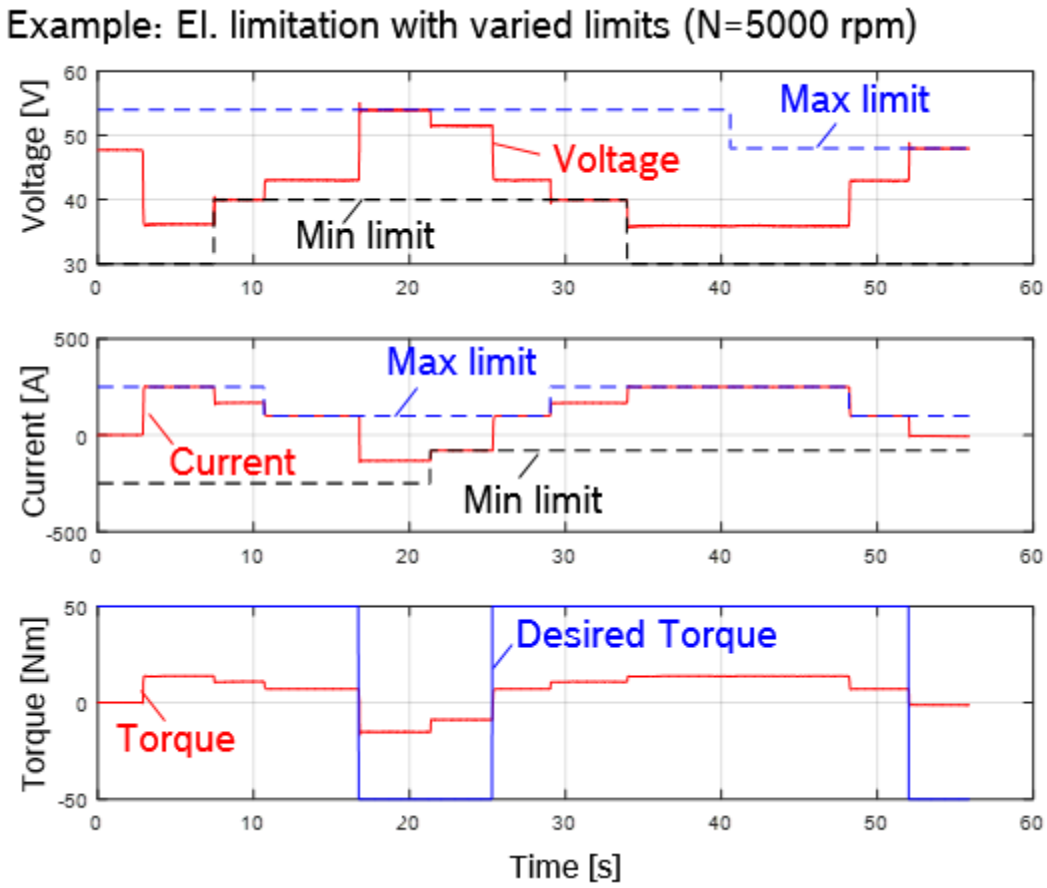
The image <Internal function schematics of voltage & current limitation> gives an overview about the internal implementation of the electrical limitation function in the BRM SW. A possible result achieved with this electrical limitation function is shown in image <Example of electrical limitation>.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Internal function schematics of voltage & current limitation>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



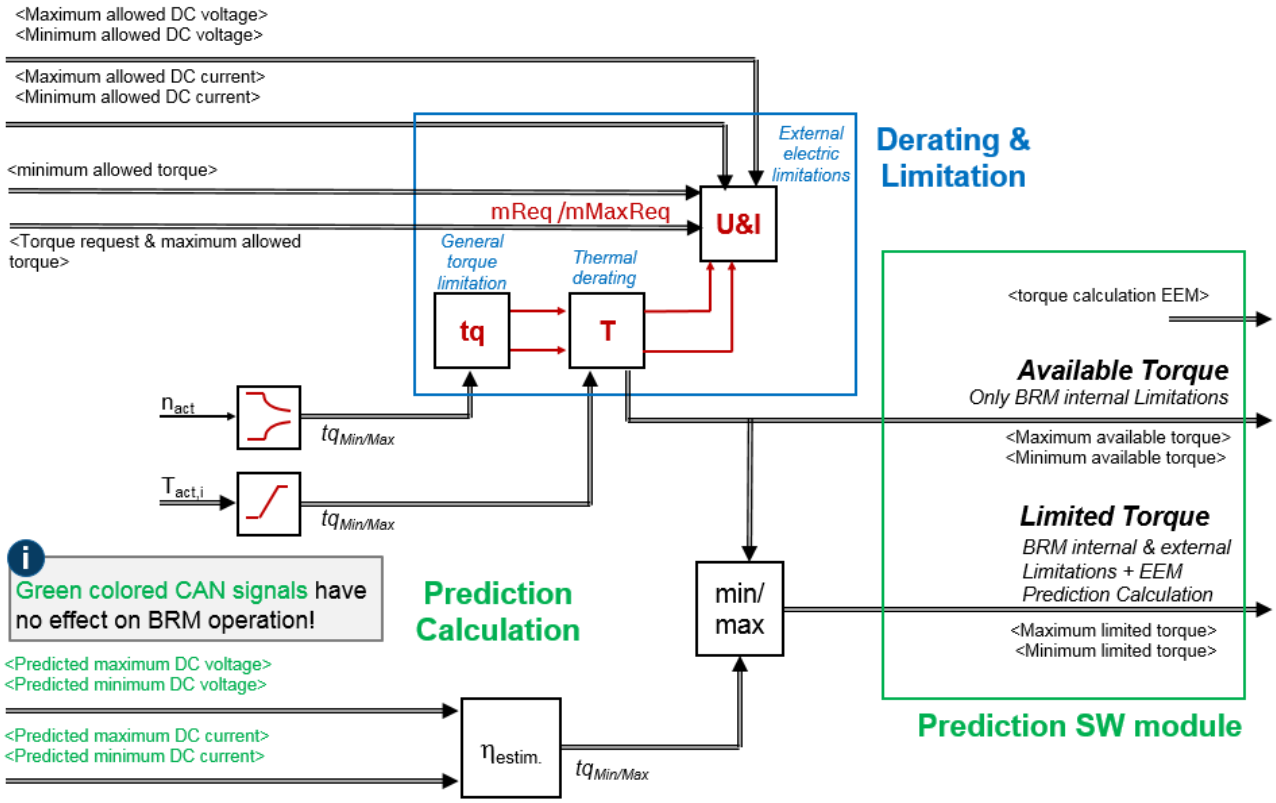
<Example of electrical limitation>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

6.7 Prediction signals & EEM calculations

Implementation of prediction in SW releases from 320_320_C_04_01 (includes all C- and D-Sample releases)

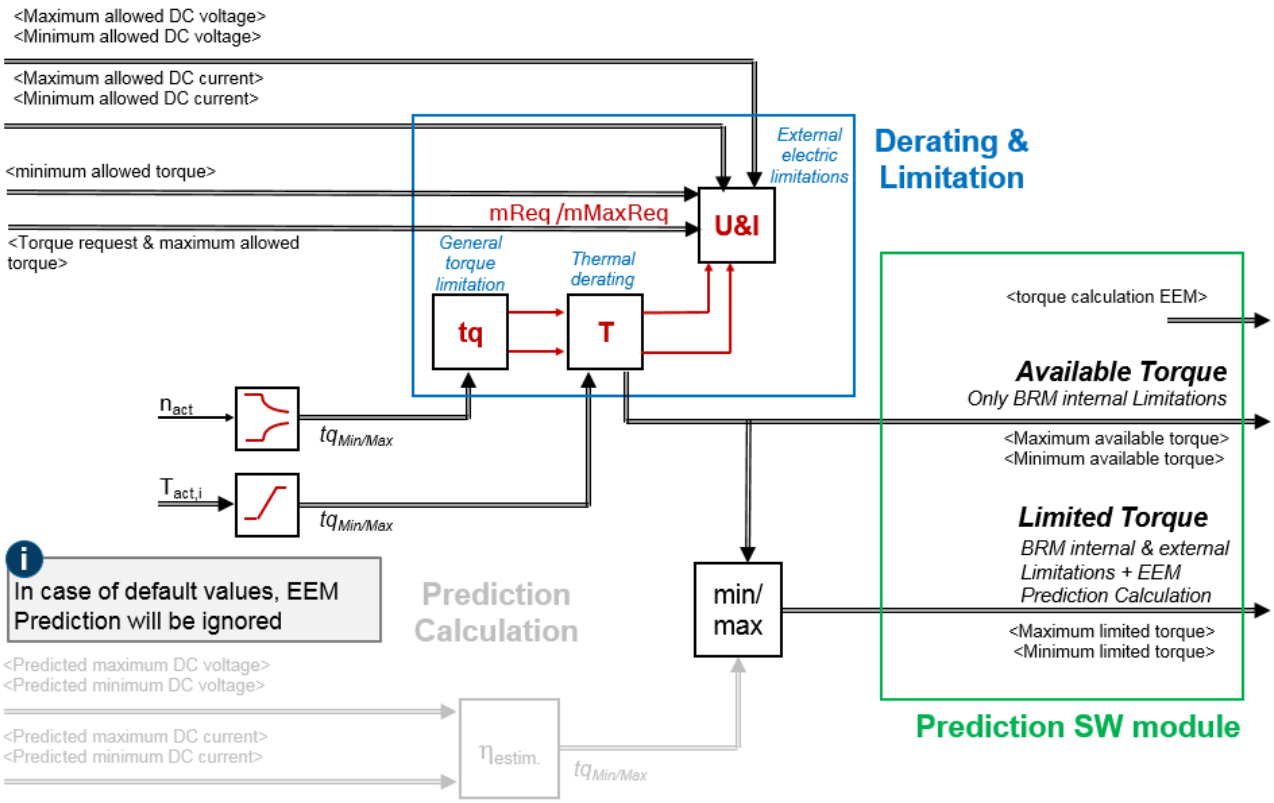
The interaction between the BRM concept for Derating, Limitation & Prediction and the CAN input/output signals is shown in the image <Advanced concept for Derating, Limitation & Prediction>.



<Advanced concept for Derating, Limitation & Prediction>

The minimum/maximum predicted torque output of the BRM is based on the internal used minimum/maximum limited torque resulting from derating & limitation (please see image <Advanced concept for Derating, Limitation & Prediction>).

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Deactivation of external prediction signals>

If the external prediction is not used in the application, the default values for EEM prediction signals shall be used (please see image <Deactivation of external prediction signals>).

Boost/Recuperation in combination with EEM Prediction

Based on the prediction values from EEM, the BRM calculates prediction values for the maximum predicted torque and returns them to PT. In this manner, the maximum predicted torque value is known already before the actual request to the BRM at any time.

The EEM continuously sends the prediction values for voltage and current in the signals:

Motor mode:

<Minimum predicted DC voltage> & <maximum predicted DC current>

Generator mode:

<Maximum predicted DC voltage> & <minimum predicted DC current>

From these values the BRM calculates the predicted torque and merges it in the signals:

Motor mode:

<Maximum limited torque>

Generator mode:

<Minimum limited torque>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

The following calculations are considered:

Motor mode:

$$maximum\ predicted\ torque = \frac{< minimum\ predicted\ DC\ voltage > * < maximum\ predicted\ DC\ current > * 0.82}{< actual\ speed >}$$

$$< Maximum\ limited\ torque > = \min(maximum\ predicted\ torque, < maximum\ limited\ torque >)$$

Generator mode:

$$minimum\ predicted\ torque = \frac{< maximum\ predicted\ DC\ voltage > * < minimum\ predicted\ DC\ current > * 0.83}{< actual\ speed >}$$

$$< Minimum\ limited\ torque > = \min(minimum\ predicted\ torque, < minimum\ limited\ torque >)$$

Calculated BRM torque prediction signals or EEM prediction values are not used for internal limitation of BRM torque. For the purpose of electric power limitation, the EEM sends specific current and voltage limits to the BRM, which will be considered in either case:

Motor mode:

<Minimum allowed DC voltage> & <maximum allowed DC current>

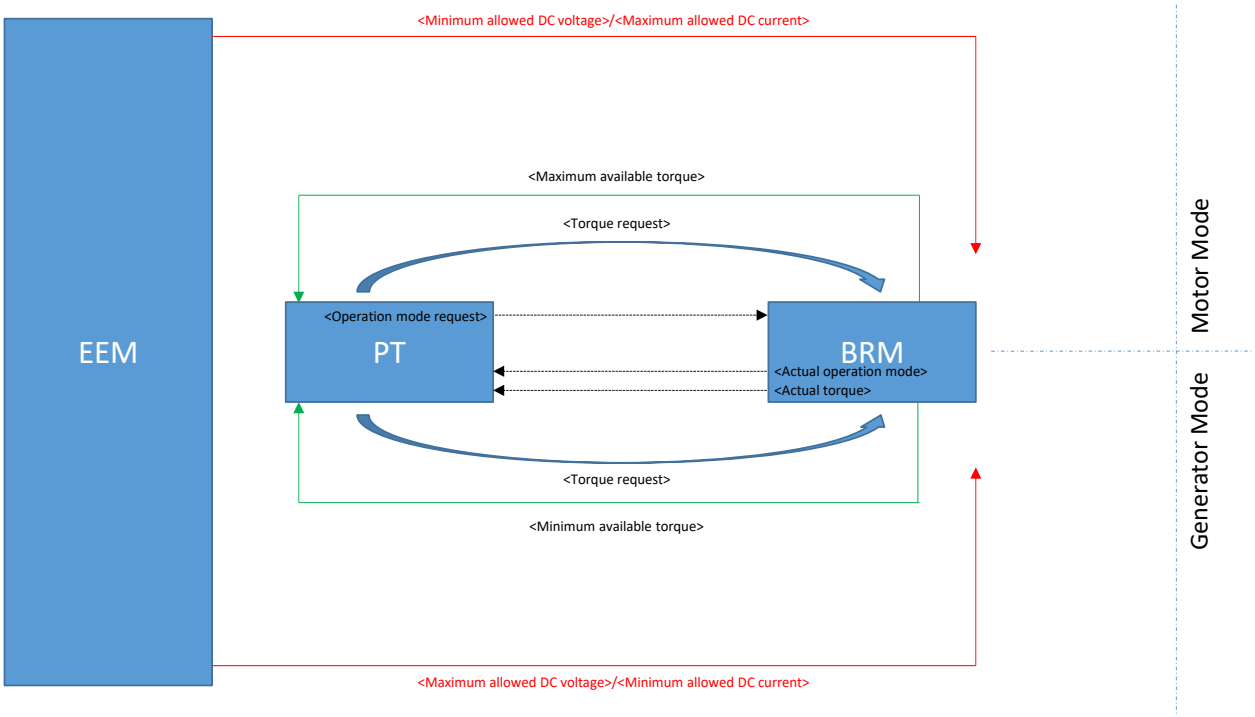
Generator mode:

<Maximum allowed DC voltage> & <minimum allowed DC current>

On the current level of development, the torque prediction values communicated by the BRM are based on simplified assumptions (i.e. roughly estimated efficiency). If a thermal derating is active, the calculated torque prediction values are limited to the derated torque values.

The signal flow is displayed in the Image below.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Control Strategy - Prediction Signals>

State Of Charge (SOC) reduction / increase – strategy

The EEM can sends a requested DC current to the BRM in the signal <current calculation EEM>. With this value the BRM calculates a mechanical power and sends it to the PT in the signals <torque calculation EEM> and <speed calculation EEM>. This torque can be considered by PT in its strategy (since torque is still requested by PT and not by EEM).

For the purpose of power limitation, the EEM sends values for DC current and DC voltage limitation to the BRM, which will be considered in either case.

The following calculation steps are considered:

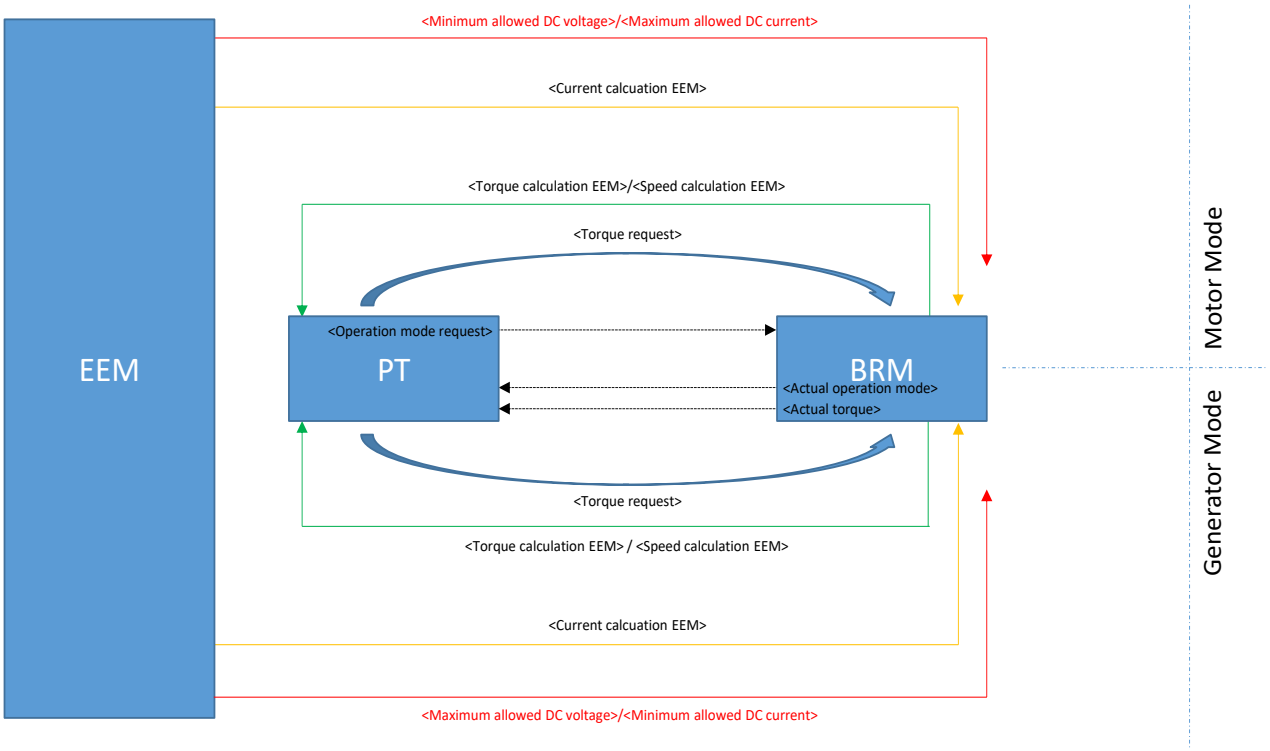
$$< torque\ calculation\ EEM > = \frac{< current\ calculation\ EEM > * < actual\ DC\ voltage >}{< actual\ speed > * 0.7}$$

$$< torque\ calculation\ EEM > within < minimum\ limited\ torque > \text{ and } < maximum\ limited\ torque >$$

$$< speed\ calculation\ EEM > = < actual\ speed >$$

The signal flow is displayed in the Image below.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.



<Control Strategy - Request Signals>

Application Hints

•Most actual SW shall be used to get the best quality and accuracy with respect to derating, limitation & prediction.

•Also actual EEM prediction signals (i.e. <Predicted maximum DC current>, <Predicted minimum DC current>, <Predicted maximum DC voltage>, <Predicted minmum DC voltage>) can't be used to limit current or voltage of the BRM. They have only calculation purpose!

•The EEM limitation signals (i.e. <Maximum allowed DC current>, <Minimum allowed DC current>, <Maximum allowed DC voltage>, <Minimum allowed DC voltage>) shall be used instead for the purpose of electric limitation.

•If the external prediction is not used in the application, the default values for EEM prediction signals (<Predicted maximum DC current> = 500A, ...) shall be used. Please refer to image <Deactivation of external prediction signals> or chapter <Communication>.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

7 Diagnostics & Error Memory

Unified Diagnostic Services (UDS) are implemented in the BRM. Therefor the BRM ECU contains an own error memory and Diagnostic Trouble Codes (DTC).

The stored errors can be read and, if needed, deleted via CAN. These operations are provided by special communication services. These services are described in specific files (*.pdx file). The *.pdx file is integrated within the CANoe configuration. It can be found in the window „BRS_Inverter – Diagnose-Console“. In the topic „Fault Memory“, the diagnosis services to read and delete the error memory entries can be found.

A recognized error is communicated by sending the signal <actual operation mode> = "Failure". This state can be terminated by a Post-Run / Initialization process. If the error cause is still present, the BRM will send again <actual operation mode> = "Failure".

Each stored error within the error memory has a state:

- "Active" – recognized in the actual cycle and active error ("DTC Statusbyte.Test failed" = true)
- "Inactive" – recognized in a former cycle and in the actual cycle no longer active error (error is "healed") ("DTC Statusbyte.Test failed" = false)

The implemented errors are listed in the following table.

This overview contains information about each error entry. Beside the error description, the test condition "Tested in" and the reaction to this error "System reactions after failure" are described.

System reactions after failure:

- "Normal mode": No operating changes will be remarked.
- "Default sensor signals": Torque output can be reduced due to default sensor values but normal operation is still possible.
- "Default CAN signals": Default CAN signal values are used, if CAN signals/messages are missing or invalid. Please see chapter <Communication >> CAN-Matrix> for detailed information.
- "FaultInverterState controlled (IDLE)": The BRM switches to <actual operation mode> = "Idle" and remains there.
- "Safe/FaultInverterState" or "Safe/FaultSystemState": The BRM switches to <actual operation mode> = "Failure" and remains there.
- "Limp home": The BRM switches to <actual operation mode> = "Idle" and remains there. Please note: This DTC will not be triggered by this SW.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

DTC	Name	Description	Tested in	System reaction after failure
1	RotP_E_TRIM	Rotor Position Sensor - Not Trimmed (electrical and mechanical).	PreRun	FaultInverterState
2	RotP_E_SCP	Rotor Position Sensor - Short Circuit Plus	Run	FaultInverterState
3	RotP_E_SCM	Rotor Position Sensor - Short Circuit Minus	Run	FaultInverterState
4	RotP_E_TLOW	Rotor Position Sensor - Radius too Low	Run	FaultSystemState
5	RotP_E_THIGH	Rotor Position Sensor - Radius too High	Run	FaultSystemState
7	RotP_E_PLAUS_TMCHA	Rotor Position Sensor - Gradient too High	Run	FaultSystemState
10	iExct_E_THIGH	Excitation Current Sensor - Range too High	Run	FaultSystemState
11	iExct_E_TLOW	Excitation Current Sensor - Range too Low	Run	FaultSystemState
12	iExct_E_FACTOR	Excitation Current Sensor - Correction factor zero	PreRun	FaultInverterState
14	iExct_E_PLAUS	Excitation Current Sensor - Deviation between measured and calculated (model based) excitation current too high	Run	FaultSystemState
26	iPha_E_THIGH_U	Phase Current Sensor - Range too High (Phase U)	Run	FaultSystemState
27	iPha_E_THIGH_V	Phase Current Sensor - Range too High (Phase V)	Run	FaultSystemState
28	iPha_E_THIGH_W	Phase Current Sensor - Range too High (Phase W)	Run	FaultSystemState
29	iPha_E_THIGH_X	Phase Current Sensor - Range too High (Phase X)	Run	FaultSystemState
30	iPha_E_THIGH_Y	Phase Current Sensor - Range too High (Phase Y)	Run	FaultSystemState
31	iPha_E_TLOW_U	Phase Current Sensor - Range too Low (Phase U)	Run	FaultSystemState
32	iPha_E_TLOW_V	Phase Current Sensor - Range too Low (Phase V)	Run	FaultSystemState
33	iPha_E_TLOW_W	Phase Current Sensor - Range too Low (Phase W)	Run	FaultSystemState
34	iPha_E_TLOW_X	Phase Current Sensor - Range too Low (Phase X)	Run	FaultSystemState
35	iPha_E_TLOW_Y	Phase Current Sensor - Range too Low (Phase Y)	Run	FaultSystemState
46	iPha_E_PLAUS_SUM	Phase Current Sensor - Sum of all phase currents not zero	Run	FaultSystemState
48	tPwr_E_DBC1_SCP	DBC1 temperature sensor - Short Circuit Plus	Run	FaultInverterState Controlled (IDLE)
49	tPwr_E_DBC1_SCM	DBC1 temperature sensor - Short Circuit Minus	Run	FaultInverterState Controlled (IDLE)
50	tPwr_E_DBC1_THIGH	DBC1 temperature sensor - Range too High	Run	FaultInverterState Controlled (IDLE)
51	tPwr_E_DBC1_TLOW	DBC1 temperature sensor - Range too Low	Run	FaultInverterState Controlled (IDLE)
53	tPwr_E_DBC2_SCP	DBC2 temperature sensor - Short Circuit Plus	Run	FaultInverterState Controlled (IDLE)
54	tPwr_E_DBC2_SCM	DBC2 temperature sensor - Short Circuit Minus	Run	FaultInverterState Controlled (IDLE)
55	tPwr_E_DBC2_THIGH	DBC2 temperature sensor - Range too High	Run	FaultInverterState Controlled (IDLE)
56	tPwr_E_DBC2_TLOW	DBC2 temperature sensor - Range too Low	Run	FaultInverterState Controlled (IDLE)
58	tPwr_E_DBC_PLAUS	DBC1 temperature sensor - Deviation between DBC1 and DBC2 too high	PreRun	FaultInverterState Controlled (IDLE)
59	tCAir_E_COOLG_AIR_SCP	Cooling Air temperature sensor - Short Circuit Plus	Run	Default sensor signals
60	tCAir_E_COOLG_AIR_SCM	Cooling Air temperature sensor - Short Circuit Minus	Run	Default sensor signals
61	tCAir_E_COOLG_AIR_THIGH	Cooling Air temperature sensor - Range too High	Run	Default sensor signals
62	tCAir_E_COOLG_AIR_TLOW	Cooling Air temperature sensor - Range too Low	Run	Default sensor signals
63	tCAir_E_COOLG_AIR_TMCHA	Cooling Air temperature sensor - Too Much Change (difference between prior and current value too high)	Run	Default sensor signals
64	tCAir_E_COOLG_AIR_PLAUS	Cooling Air temperature sensor - Deviation between outside air temperatur (after vehicle off time) and cooling air temperature too high	PreRun	Default sensor signals
65	tLogic_E_LTCC_SCP	LTCC temperature sensor - Short Circuit Plus	Run	FaultInverterState Controlled (IDLE)
66	tLogic_E_LTCC_SCM	LTCC temperature sensor - Short Circuit Minus	Run	FaultInverterState Controlled (IDLE)
67	tLogic_E_LTCC_THIGH	LTCC temperature sensor - Range too High	Run	FaultInverterState Controlled (IDLE)
68	tLogic_E_LTCC_TLOW	LTCC temperature sensor - Range too Low	Run	FaultInverterState Controlled (IDLE)
69	tLogic_E_LTCC_TMCHA	LTCC temperature sensor - Too Much Change (difference between prior and current value too high)	Run	FaultInverterState Controlled (IDLE)
70	tLogic_E_LTCC_PLAUS	LTCC temperature sensor - Deviation between DBC temperature and LTCC temperature too high	PreRun	FaultInverterState Controlled (IDLE)

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

DTC	Name	Description	Tested in	System reaction after failure
71	SplyV_E_UB48_SLOW_SCM	Supply Voltage Sensor UB48SLOW - Short Circuit Minus	Run	FaultInverterState
72	SplyV_E_UB48_SLOW_SCP	Supply Voltage Sensor UB48SLOW - Short Circuit Plus	Run	FaultSystemState
73	SplyV_E_UB48_SLOW_THIGH	Supply Voltage Sensor UB48SLOW - Range too High	Run	FaultSystemState
74	SplyV_E_UB48_SLOW_TLOW	Supply Voltage Sensor UB48SLOW - Range too Low	Run	FaultInverterState
75	SplyV_E_UB48_PLAUS	Supply Voltage Sensor UB48 - Deviation between UB48 Slow voltage and UB48 Fast voltage too high	Run	FaultSystemState
76	SplyV_E_UB48_OV	Supply Voltage Sensor UB48 - Overvoltage	Run	Normal mode
77	SplyV_E_UB48_UV	Supply Voltage Sensor UB48 - Undervoltage	Run	FaultInverterState Controlled (IDLE)
78	SplyV_E_UB48_FAST_SCM	Supply Voltage Sensor UB48FAST - Short Circuit Minus	Run	FaultInverterState
79	SplyV_E_UB48_FAST_SCP	Supply Voltage Sensor UB48FAST - Short Circuit Plus	Run	FaultSystemState
80	SplyV_E_UB48_FAST_THIGH	Supply Voltage Sensor UB48FAST - Range too High	Run	FaultSystemState
81	SplyV_E_UB48_FAST_TLOW	Supply Voltage Sensor UB48FAST - Range too Low	Run	FaultInverterState
82	SplyV_E_UBS_SCM	Supply Voltage Sensor UBS - Short Circuit Minus	Run	FaultSystemState
83	SplyV_E_UBS_SCP	Supply Voltage Sensor UBS - Short Circuit Plus	Run	FaultSystemState
84	SplyV_E_UBS_THIGH	Supply Voltage Sensor UBS - Range too High	Run	FaultSystemState
85	SplyV_E_UBS_TLOW	Supply Voltage Sensor UBS - Range too Low	Run	FaultSystemState
86	SplyV_E_UBSR11_SCM	Supply Voltage Sensor UBSR11 - Short Circuit Minus	Run	FaultSystemState
87	SplyV_E_UBSR11_SCP	Supply Voltage Sensor UBSR11 - Short Circuit Plus	Run	FaultSystemState
88	SplyV_E_UBSR11_THIGH	Supply Voltage Sensor UBSR11 - Range too High	Run	FaultSystemState
89	SplyV_E_UBSR11_TLOW	Supply Voltage Sensor UBSR11 - Range too Low	Run	FaultSystemState
90	SplyV_E_UBSR11_UBS_PLAUS	Supply Voltage Sensor UBSR11 - Deviation between UBSR11 and UBS too high	Run	FaultSystemState
91	SplyV_E_UB12_SCM	Supply Voltage Sensor UB12 - Short Circuit Minus	Run	FaultSystemState
92	SplyV_E_UB12_SCP	Supply Voltage Sensor UB12 - Short Circuit Plus	Run	FaultSystemState
93	SplyV_E_UB12_THIGH	Supply Voltage Sensor UB12 - Range too High	Run	FaultSystemState
94	SplyV_E_UB12_TLOW	Supply Voltage Sensor UB12 - Range too Low	Run	FaultSystemState
95	SplyV_E_UB12_UBS_PLAUS	Supply Voltage Sensor UB12 - Deviation between UB12 and UBS too high	PreRun	FaultSystemState
99	OutStC_E_HW_ERROR	Outputstage control: Bridge driver switched off permanent because of HW error	Run	SafeInverterState
105	Com_E_TOUT_Eem_Edrv_Des_1	Message Timeout	Run	Default CAN signals
106	Com_E_CRC_Eem_Edrv_Des_1	CRC Error	Run	Default CAN signals
107	Com_E_SQC_Eem_Edrv_Des_1	Sequence / Alive Counter Error	Run	Default CAN signals
108	Com_E_TOUT_Eem_Edrv_Lim_1	Message Timeout	Run	Default CAN signals
109	Com_E_CRC_Pt_Edrv_Des_1	CRC Error	Run	Default CAN signals
110	Com_E_SQC_Pt_Edrv_Des_1	Sequence / Alive Counter Error	Run	Default CAN signals
111	Com_E_TOUT_Pt_Edrv_Des_1	Message Timeout	Run	Default CAN signals
113	Com_E_TOUT_Pt_Veh_1	Message Timeout	Run	Default CAN signals
115	CANIF_E_INVALID_DLC	internal SW error	Run	FaultInverterState
118	CANSM_E_BUSOFF_NETWORK	internal SW error	Run	FaultInverterState
122	PDUR_E_PDU_INSTANCE_LOST	internal SW error	Run	FaultInverterState
126	NvM_E_READ_FAILED	internal SW error	Run	Normal mode
127	NvM_E_WRITE_FAILED	internal SW error	Run	Normal mode
129	NvM_E_INTEGRITY_FAILED	internal SW error	Run	Normal mode
131	Com_E_TOUT_Pt_Obd_1	Message Timeout	Run	Normal mode
133	Com_E_TOUT_Pt_Obd_2	Message Timeout	Run	Normal mode
135	SPI_E_TRANSFER_TIMEOUT	internal SW error	Run	FaultInverterState
136	SPI_E_DMA_TRANSFER	internal SW error	Run	FaultInverterState
137	SPI_E_RECEIVE_OVERFLOW	internal SW error	Run	FaultInverterState
138	SPI_E_TRANSFER_FIFO	internal SW error	Run	FaultInverterState
139	SPI_E_TRANSMIT_UNDERFLOW	internal SW error	Run	FaultInverterState

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

DTC	Name	Description	Tested in	System reaction after failure
142	CAN_E_TIMEOUT	internal SW error	Run	FaultInverterState
149	SyAsic_E_SPI_TOUT	SPI Timeout System ASIC CL260	Run	FaultInverterState
151	LdDmpC_E_CIRCUIT_LD_DMP	load dump comparator circuit not working	PreRun	FaultInverterState
152	LdDmpC_E_ACTIVE_LD_DMP	load dump situation detected	Run	FaultSystemState
165	SyAsic_E_CONFIG	System ASIC "configuration error"	PreRun	FaultInverterState
167	BattS_E_CUTOFF_SWT	Battery Sensor "Battery cutoff switch open"	Run	Limp home
168	LdDmpC_E_CIRCUIT_GND_LOSS	ground loss comparator circuit not working	PreRun	FaultInverterState
169	LdDmpC_E_ACTIVE_GND_LOSS	ground loss situation detected	Run	FaultInverterState
170	Os_E_LIMIT_CORE_0	RunTime violation detected in OS (Core 0)	Run	Normal mode
171	Os_E_GENERAL	General Error detected in OS	Run	Normal mode
172	tLogic_E_LTCC_TLCHA	LTCC temperature sensor - Too Less Change in time window	Run	FaultInverterState Controlled (IDLE)
173	tCAir_E_COOLG_AIR_TLCHA	Cooling Air temperature sensor - Too Less Change in time window	Run	Default sensor signals
188	uExctC_E_PLAUS_PWM	Excitation Voltage Coordinator - target and measured PWM not plausibel	Run	FaultInverterState
189	uPhaC_E_PLAUS_PWM	Phase Voltage Coordinator - target and measured PWM not plausibel	Run	FaultInverterState
191	iPha_E_FACTOR_Y	Phase Current Sensor - Correction factor zero	Run	FaultInverterState
192	SyFctC_E_UB12_SWITCH	Functionality of UB12_Disable switch	Run	FaultInverterState
193	SyFctC_E_UBSR11_SWITCH	Functionality of UBSR_Enable switch	Run	FaultInverterState
194	tPwr_E_DBC1_CD	DBC1 temperature sensor - Circuit discontiunity	Run	FaultInverterState Controlled (IDLE)
195	tPwr_E_DBC2_CD	DBC2 temperature sensor - Circuit discontiunity	Run	FaultInverterState Controlled (IDLE)
196	tLogic_E_LTCC_CD	LTCC temperature sensor - Circuit discontiunity	Run	FaultInverterState Controlled (IDLE)
197	tCAir_E_COOLG_AIR_CD	Cooling Air temperature sensor - Circuit discontiunity	Run	Default sensor signals
200	Os_E_LIMIT_CORE_1	RunTime violation detected in 2nd core (Core 1)	Run	Normal mode
201	RotP_E_PLAUS_OFFSET_TLOW	Rotor Position Sensor - Offset Error rotor position sensor too low	Run	FaultSystemState
202	iExct_E_PLAUS_OFFSET_THIGH	Excitation Current Sensor - Offset deviation too High	Run	FaultSystemState
203	iPha_E_FACTOR_U	Phase Current Sensor - Correction factor zero	Run	FaultInverterState
204	iPha_E_FACTOR_V	Phase Current Sensor - Correction factor zero	Run	FaultInverterState
205	iPha_E_FACTOR_W	Phase Current Sensor - Correction factor zero	Run	FaultInverterState
206	iPha_E_FACTOR_X	Phase Current Sensor - Correction factor zero	Run	FaultInverterState
207	OutStC_E_SYS_ERROR	Outputstage control: Bridge driver switched off permanent (system error)	Run	FaultInverterState
209	RotP_E_PLAUS_HALL	Rotor Position Sensor - Plaus of digital Hall signal and RotP	Run	Normal mode
211	Com_E_CRC_Pt_Veh_1	CRC Error	Run	Default CAN signals
212	Com_E_SQC_Pt_Veh_1	Sequence / Alive Counter Error	Run	Default CAN signals
213	SysDat_E_VARIANT_INV	System Data - gear type resp. OBD variant read from NvM is invalid	Startup	FaultInverterState
214	iExct_E_PLAUS_OFFSET_TLOW	Excitation Current Sensor - Offset deviation too Low	Run	FaultSystemState
215	ShDwnU_E_L2_TRQ	ShutDownUnit: TRQ test failed	Run	SafeInverterState
216	ShDwnU_E_L2_CRASH	ShutDownUnit: CRASH test failed	Run	SafeSystemState
217	ShDwnU_E_L2_VOLT	ShutDownUnit: VOLT test failed	Run	SafeSystemState
218	ShDwnU_E_L2_MON	ShutDownUnit: L2 MON test failed	Run	SafeInverterState
219	ShDwnU_E_L3_MON	ShutDownUnit: L3 MON test failed	Run	SafeInverterState
220	RotP_E_PLAUS_OFFSET_THIGH	Rotor Position Sensor - Offset Error rotor position sensor too high	Run	FaultSystemState
221	SyFctC_E_UBS_SWITCH	Functionality of UBS_Enable switch	Run	FaultInverterState
222	Tq_E_PRFMNC_GEN_COLDSTART	Torque Generator during engine coldstart	Run	Normal mode
223	Tq_E_PRFMNC_GEN_NO_COLDSTART	Torque Generator after engine coldstart	Run	Normal mode
224	ElPwr_E_PRFMNC_GEN_COLDSTART	Peformance Generator during engine coldstart	Run	Normal mode
225	ElPwr_E_PRFMNC_GEN_NO_COLDSTART	Peformance Generator after engine coldstart	Run	Normal mode
226	Tq_E_PRFMNC_MOT_COLDSTART	Torque Motor during engine coldstart	Run	Normal mode
227	Tq_E_PRFMNC_MOT_NO_COLDSTART	Torque Motor after engine coldstart	Run	Normal mode
228	ElPwr_E_PRFMNC_MOT_COLDSTART	Performance Motor during engine coldstart	Run	Normal mode
229	ElPwr_E_PRFMNC_MOT_NO_COLDSTART	Performance Motor after engine coldstart	Run	Normal mode
230	SplyV_E_UB12_THIGH_JS	Supply Voltage Sensor UB12 - Range too High (Jump Start)	Run	FaultSystemState
231	Com_E_TOUT_Eem_Edrv_Pred_1	Message Timeout	Run	Default CAN signals
232	Com_E_TOUT_Obd_state_1	Message Timeout	Run	Default CAN signals

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

8 SW Update (Flash)

The software running on our different samples is subject of continuous development, both in terms of the implementation of new/improved functionality, as well as in terms of bug fixing. In case a new software update has been released for this hardware variant, the BRM can be flashed in the way described below.

Preconditions:

The following components are needed:

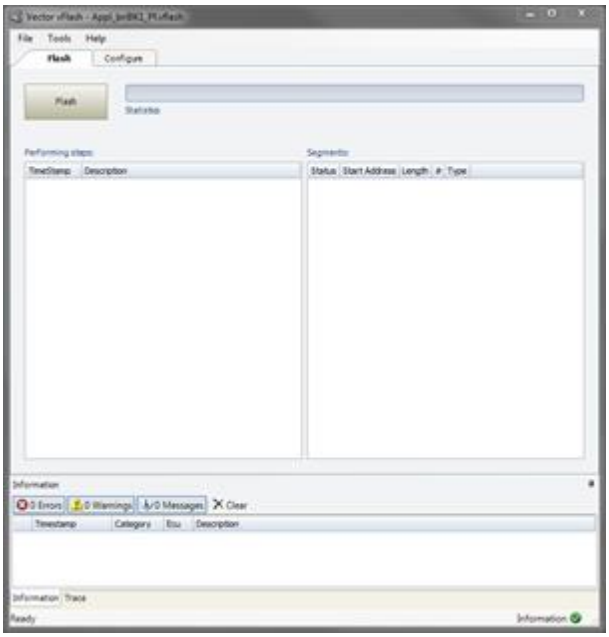
- A suitable hardware & software for CAN communication (i.e. CANcase & CANoe) and a host PC
- Vector software vFlash (Version 3.1 SP2 or later) & corresponding licenses

The SW update will be provided in a folder called "vFlash", which contains all the relevant BRM software files. Alternatively a compressed "vflashpack" file is provided.

After installing vFlash, the Vector hardware has to be pre-configured to be ready to use with vFlash. Please check in menu "Vector Hardware" if the correct hardware and channel is assigned.

Flash process preparation:

- A robust 12 V supply has to be applied before starting a flash procedure.
- The CAN communication to BRM has to be established.
- The signal <ignition switch state> shall be "on" and the signal <operation mode request> shall be "Idle".
- BRM shall respond <actual run state> = "Pre-Run" or "Run" and <actual operation mode> "Standby" or "Idle".
- BRM rotation speed shall be zero.
- There is no dependency on the voltage applied to the 48V circuit.



<Screenshot Vector vFlash>

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

Flashing

CAN communication to BRM shall not be stopped (CANoe, dSpace, test bench, etc.) in order to avoid a sleep state of CAN Transceiver. Run the file "*.vflashpack" provided by SEG Automotive. Vector vFlash will open as shown in the image <Screenshot Vector vFlash>. By clicking on the panel "Flash", the flash process will be started. The supply voltage and the communication shall not be interrupted during the flash procedure. As soon as the update procedure has finished, a specific text will be displayed in the window "Performing steps".

The diagnostic services "Extended Session Start" and "Programming Session Start" are requested automatically by vFlash and don't have to be sent manually.

If flashing ends successfully BRM shall be restarted by switching off/on 12 V and 48 V supply.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.

9 Known Bugs

Unsolved problems in the software 320_320_D_01_01:

- Initialization at speed is not working at n = 100 rpm.
- BRM sticks in PreRun if Ignition is reset at -30°C climate temperature.
- BRM goes to ErrorMode sporadically after Ignition off/on.

© SEG Automotive Germany GmbH 2018. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for the industrial property rights.